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Investigating the correlation of biometric and sensory data for fall detection using deep learning models

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Abstract

Background: Falls are a major threat to health and independence, especially in older persons. Although sensor-based fall detection systems are common, their performance can be greatly affected by individual physiological and demographic features. This research will examine how such individual biometric data affects fall detection accuracy.

Objective: This research specifically aimed to increase the accuracy of fall detection by combining biometric and sensor information, using deep learning models trained and tested on the MobiAct dataset.

Methods: To assess the effect of biometric and demographic information on the performance of fall detection, we carried out six incremental experiments. In every experiment, more biometric features (gender, height, age, weight) were added systematically to a sensor dataset. Stacked long short-term memory (LSTM) and gated recurrent unit (GRU) networks were utilized as the deep learning models for analysis because they effectively handle sequential sensor data.

Results: In Experiment 1 with only the sensory data, maximum performance was seen, with the stacked GRU performing at 96% average accuracy, precision, recall, and F1-score, performing better than the Stacked LSTM. Experiments 2 (age included) and 4 (gender included) always gave 95% average accuracy for the GRU, and Experiment 5 (weight included) achieved a 93% average. Stacked GRU remained strong even when all the biometric features were stacked together (Experiment 6, 91% accuracy), while the performance of the stacked LSTM was more impacted by intricate multimodal inputs.

Conclusions: The results of our investigation highlight that personal biometric information can play a major role in improving fall detection rates and optimizing model performance, even when considered for a dataset of young adults who simulated falls. The findings reiterate the significance of individualized methods in fall detection systems and provide foundational underpinnings for subsequent investigations aimed at older adult populations with more varied and representative data.

Keywords

RNNs, LSTM, gated recurrent unit networks, automatic fall detection systems, micro-electromechanical system, deep learning

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Introduction

The rising population aging globally has profoundly increased the incidence and influence of falls in older individuals, making them a central public health issue. Falls are sudden, unplanned changes in body position that bring individuals into contact with the ground, often resulting from age-related physiological decline in neuromuscular control, proprioception, and vestibular function.¹ Falling is also significantly prevalent in secluded indoor settings, where prompt intervention is likely to be prohibitively delayed. Epidemiological studies suggest that nearly one-third of adults over the age of 65 have at least one fall each year, and falls are the second highest cause of accidental deaths in this group.² In addition to such immediate physical injuries as fractures and traumatic brain injury, falls also cause severe psychological harm, including anxiety syndrome after a fall, enhanced social isolation, and a significant decrease in overall quality of life. Such synergistic morbidity and mortality firmly emphasize the imperative necessity of sophisticated technological solutions to provide early detection and intervention.

The acute problem of falls has driven research on automatic fall detection systems (AFDSs). Traditionally, AFDS methods have largely fallen into the vision-based and wearable sensor-based types. Vision-based systems, which utilize advanced computer vision algorithms to examine video streams for falls occurrence, provide benefit in terms of non-contact monitoring and rich contextual information. But their real-world utilization is greatly limited by intrinsic drawbacks. These encompass ubiquitous privacy issues resulting from ongoing video surveillance, sensitivity to changing lighting and occlusions, and high computational requirements and infrastructure expenses of processing high-resolution video.^{3–6} Therefore, these weaknesses make vision-based systems practically impossible for common, ongoing deployment in privacy-sensitive settings such as personal residences or care facilities.

By contrast, wearable sensor-based AFDS have evolved as a strong, privacy-respecting, and inexpensive solution. Such systems commonly use micro-electromechanical system (MEMS) inertial sensors, mostly tri-axial accelerometers and gyroscopes, complemented with barometers, integrated into pervasive personal devices (e.g., smartphones, smartwatches) or special purpose wearable patches.⁷ This solution allows for ambient, unobtrusive monitoring without dependency on surrounding illumination or line-of-sight requirements, thus practically addressing the privacy issues related to camera-based surveillance. The intrinsic portability, cost-effectiveness, and compatibility with current wireless communications infrastructures make wearable sensor-based systems flexible and scalable to a wide range of clinical and home-based applications.

The initial versions of wearable AFDS were mostly based on threshold-based detection methods.^{8,9} Such methods work

by comparing specific extracted features from sensor signals (e.g., magnitude of the acceleration, angular velocity) against empirically derived threshold values to categorize an event as a fall. Although computationally light and easy to deploy, threshold-based systems have severe drawbacks. Their strict, pre-defined thresholds frequently are not able to estimate the innate inter-subject variation in human gait, motion patterns, and fall dynamics, resulting in high false positive (e.g., discriminating a fall from daily living activities such as sitting quickly) and false negative (missed falls) rates.¹⁰ In addition, their failure to dynamically adjust themselves in response to shifting environmental conditions or personal physiological statuses severely limits their robustness and generalizability. While computationally less expensive, their accuracy and reliability are always overshadowed by data-driven machine learning (ML) methods.

The following development of AFDS integrated conventional machine learning (ML) models, which call for a two-step process: manual feature extraction and classification.¹¹ This entails domain specialists carefully hand-crafting characteristics like signal peaks, root mean square (RMS) values, or frequency domain features from sensor raw data. Although providing enhanced discriminatory power compared to threshold-based approaches, this feature engineering process is time-consuming, manpower-intensive, and strongly reliant on the proficiency and experience of human operators. In addition, conventional ML algorithms tend to have a hard time with the size, high dimensionality, and temporal dependence found in large-scale, multi-sensor time-series datasets, especially when required to aggregate data across various sensing modalities effectively.

Deep learning (DL) has recently transformed time-series analysis by solving the challenges posed by manual feature engineering. DL models like recurrent neural networks (RNNs) e.g., long short-term memory (LSTM) and gated recurrent unit (GRU) networks can learn hierarchical, discriminative features independently from raw sequential input data. The end-to-end learning ability greatly simplifies the model-building process, improves the flexibility, and also enhances the performance of pattern recognition tasks that are complex in nature. As a result, DL has become the cutting-edge solution to many real-world problems, such as human activity recognition and fall detection (FD). Notwithstanding these impressive advances, one enduring and important problem in the area remains the significant minimization of false fall alarms while at the same time retaining or even boosting high detection performance, especially in heterogeneous and unconstrained scenarios.

FDSs are generally classified into two fundamental approaches: Vision-based and wearable sensor-based. Vision-based systems utilize cameras to observe people and identify falls based on visual inspection. Although they have real-time functionality, privacy issues resulting from continuous monitoring severely limit their application in private and sensitive settings. In addition, their effectiveness is

routinely disrupted in poor-light environments or where visual obstructions take place, further reducing their operational utility.

Conversely, wearable sensor-based solutions employ small devices that have sensors like accelerometers, gyroscopes, and micro switches integrated within them. These are generally placed around the waist, chest, or wrist and record minute movement data to detect anomalies corresponding to a fall.^{4,12–16} Wearable systems offer a promising, privacy-friendly solution compared to their vision-based counterparts. By gathering data either near real-time or in real-time, these sensors support continuous analysis and are very versatile to operate in different environmental conditions, such as under poor lighting.

New developments in wearable FD also include the work done by Kim et al.,¹⁷ who proposed a FD methodology utilizing multi-horizon forecasting. Their work extended this idea by postulating a customized pre-prediction technique that combines both biometric and behavior-specific data of individual wearers. They utilized a temporal fusion transformer framework to fine-tune multi-horizon prediction for this task and tested its performance using F1-score, precision, recall, and accuracy on datasets like SmartFall, Notch, and DLR. Likewise, Luna-Perejón et al.¹⁸ also suggested a new wearable FD strategy that utilized RNNs. This selection complies with the recent trend of using complex machine learning methodologies for improving FD suitable for modeling sequential data. Their approach entailed training RNNs on sensor information from the SisFall dataset, assessing its contribution with F1-score, precision, recall, and accuracy.

The research by Qian et al.¹⁹ focuses on the development of a real-time wearable FDS within the context of the internet of things (IoT). Their methodology likely involves the integration of sensors with IoT technology for real-time FD. Unfortunately, the specific dataset used and the evaluation parameters are not provided in the information available. Accuracy, sensitivity, and specificity are the evaluation criteria utilized in this study. They constructed their own dataset by recruiting 29 volunteers. For elder tracking and FD, utilizing smart tiles is an innovative system presented by Daher et al.²⁰ The methodology likely involves the deployment of sensor-equipped smart tiles to monitor movements and detect falls among elderly individuals. Sensors used in this research are a force sensor and an accelerometer sensor. They developed their own dataset for this research. Evaluation criteria employed in the study, including accuracy, sensitivity, and specificity, are likely to be shared measures in determining the performance of the system for FD. One issue is that this system is effective in a restricted setting.

The study of Wu et al.²¹ investigates autonomous FD using DL with mobile sensors. The process probably includes training DL algorithms, i.e., GRU and LSTM, from mobile sensor data to detect fall events. The particular data sets used in the present study are MobiAct and

Smartwatch. Evaluation metrics used for assessing this study are sensitivity, specificity, accuracy, precision, and F1 score. Yet another new method to improve FDS by extracting an exclusive feature set from accelerometer sensor data is put forward by Le et al.²² The process probably involves gathering and processing accelerometer data to determine unique features with the purpose of enhancing FD accuracy. The particular dataset utilized in this study is MobiAct and UP Fall, which are publicly accessible. F1-score is utilized for result evaluation.

The study by Al-Rakhami et al.²³ proposes “FallDeF5,” a new FD approach utilizing 5G-based deep GRU networks. The method probably entails creating and training deep GRU networks through 5G connectivity for FD. The dataset used for this study is Smartwatch and Smartfall. Parameters for evaluation in this study may include accuracy, precision, and recall. Another accelerometer data-based human FD technique employing convolutional neural networks (CNNs) is suggested by Santos et al.²⁴ The approach most probably employs training CNN models on accelerometer data to learn patterns involved with falls. The particular datasets employed in this research are UR Fall, SmartWatch, and Notch. The evaluation criteria in this research could be accuracy, sensitivity, specificity, and precision.

The research by Wang et al.²⁵ presents “CMFALL,” a system for the identification of falls in several states that makes use of data collected by waist-mounted tri-axial accelerometers. CMFALL initially excludes non-fall data using human fall attributes with minimal computation. Following the identification of potential falls, the data is sent to a sophisticated classifier for final judgment, ensuring precise detection. A lightweight self-attention CNN is utilized as the advanced classifier to balance recognition performance and compute load, replacing the need for a deep and complicated model that demands more power and storage space. The publicly available dataset employed in this study is SiS Fall. Evaluation parameters for this research likely include accuracy, precision, recall, and the F1 score. The research by Fanez et al.²⁶ explores the integration of generalized and user-centered models for FD. The methodology likely involves combining personalized user models with generalized models to improve the accuracy of FDSs. This FD method consists of three stages: event detection, one-class problem classification to label events, and two-class classification to classify events as either fall or normal. The specific dataset used in this research is UMA Fall. Evaluation parameters in this study are expected to include sensitivity, specificity, and area under the ROC curve.

Nguyen et al. 2023²⁷ studied the Privacy-Preserving Approach to FD using sensor fusion. In this research, they introduced an integrated IoT-based system for elderly care, focusing on FD through a combination of wearable and non-wearable sensors. To address privacy concerns associated with video monitoring, they proposed a novel approach combining human skeleton extraction with

reversible visual data-hiding, ensuring data security while maintaining monitoring capabilities. Parashar et al. 2023²⁸ utilized data preprocessing and feature engineering for Gait Recognition and compared the results using ML and DL techniques. They studied the performance of deep DL and ML in gait recognition under changing conditions. From existing literature and DL pipelines, they learned popular strategies, issues, and directions related to future research. The aim is to inform researchers about data preprocessing and model choice to build effective gait recognition systems. Alawneh et al. 2023²⁹ utilized identified mobile device users based on DL and motion sensors. This work solves the problem of user identification on public devices using human activity recognition from mobile sensor readings. They introduced DL models using LSTM for user identification from movement patterns. Their experiments on publicly available datasets prove the viability of precise user identification with 90% accuracy, and they show the promise of accelerometer data compared to gyroscope data in this regard. Gharghan et al. 2024³⁰ employed AI methods for detection of the Fall for the Elderly using wireless sensor systems data. They showcased an in-depth analysis of FDSs that are tailored towards elderly individuals. It contrasts the conventional FDSs with AI-based FDSs and stresses the high performance of DL techniques. By understanding challenges, shortcomings, and possible alternatives, their work seeks to inform scholars on how to create more precise and trustworthy FDSs for the aged. Bhavani et al. 2024³¹ employed inception-based Deep CNN (DCNN) for FD and classification. Falls are an important health hazard for the aged, resulting in serious injuries and great economic loss. This paper suggests a new DL-based fall classification model, INDCNN-FDC, by integrating image preprocessing methods with Inception v3 and DCNN for efficient fall classification. Experimental outcomes compare with current techniques to show the superiority of the proposed model. Mohan et al. 2024³² used AI and IoT to identify Fall prevention for elderly. Their work discusses AI-powered fall prevention technologies, i.e., IoT, sensor, RADAR, and IR-based systems. The results show the capabilities of AI-IoT solutions for on-going, independent monitoring of older adults, lowering the risk of falls and their associated fallacies.

Kaur et al. 2024³³ identified Real-time Fall using Hybrid Haar Cascade classifier. This study presents an economical FDS with a Raspberry Pi camera and a hybrid Haar Cascade model. By precisely tracking human motion and identifying falls, the system seeks to provide greater protection to elderly people through early warnings to family members. The system proves to show better performance than current techniques in that it has an accuracy of 89.21%.

Current developments in FDSs have been propelled by the use of artificial intelligence. To improve system predictability and accuracy, researchers have investigated the addition of more parameters, including biometric data and multiple fall activities, as opposed to classical sensory

data. FD data tends to present complex, nonlinear patterns that classical methods are not very good at correctly specifying. Existing works have typically used limited activities of fall for detection and conventional machine learning approaches. This work explores the possibility of exploiting biometric attributes (age, gender, weight, height) along with multiple fall activities for enhanced performance of the FD models. With the combination of these heterogeneous sources of data, we created more effective and personalized FDSs. We lay out empirical evidence in favor of this strategy. We applied and optimized the stacked LSTM and Stacked GRU models to enhance the Fall activity detection.

Gap in knowledge

State-of-the-art FDSs in current use are largely based on analysis of general movement patterns obtained purely from inertial sensor data (accelerometers, gyroscopes). Although good for simple activity recognition, such systems rarely take into account the essential role of integrating biometric information, including age, weight, height, and gender, into their predictive models. This neglect restricts the ability of current systems to deliver genuinely personalized fall risk assessment and intervention. The generic nature of models learned on movement patterns does not consider the individual-specific physiological and biomechanical properties of people, which have significant impacts on their gait stability, falling dynamics, and recovery systems.

Precisely, there is a huge knowledge gap in understanding the systematic quantification and correlation of how multimodal fusion of various biometric parameters with multimodal sensory data exactly improves the classification robustness and accuracy of fall events through cutting-edge DL architectures. Previous work mostly reports stand-alone applications or describes qualitatively the possible advantages but does not conduct a systematic, empirical study on the individual and combined effect of separate biometric parameters on the performance and generalizability of DL-based AFDS over a wide range of fall activities (e.g., forward fall, backward fall, lateral fall, sit-to-fall, etc.). The discrepancies in reported performance in different studies are frequently explainable by this absence of personalization, and the importance of a unifying approach that takes advantage of individual-specific attributes to improve predictive models.

Purpose and objective

In addressing the knowledge gap identified, this study follows a multi-faceted aim: The main goal is to thoroughly research and quantitatively illustrate the improved performance of DL models for FD by correlating and merging extensive biometric attributes (precisely, gender, age,

weight, and height) with time-series inertial sensor data over a wide variety of simulated falling activities.

Specifically, the present study aims:

To design and fine-tune sophisticated DL architectures, namely stacked GRU and LSTM models, that can learn sophisticated spatio-temporal features from fused biometric and sensory data.

To empirically measure the incremental increase in fall classification performance (e.g., accuracy, precision, recall, F1-score, AUC) gained by using biometric data versus models learned from sensory data alone.

To examine the contribution of single biometric parameters to model performance, determining their relative importance in FD personalization.

To assess the robustness and generalizability of the introduced multimodal DL models over a wide range of various fall types and activities of daily living (ADLs), providing accurate discrimination and low false alarm rates.

By means of this study, we hope to establish sound empirical evidence in favor of incorporating biometric data in the development of more accurate, personalized, and reliable FDSs and, in the process, contribute to the state-of-the-art in elderly care technology.

Material selection: search strategy and inclusion criteria

This study is based on an extensive literature review that is guided by a systematic selection procedure, using principles often utilized in systematic review approaches. Although not an official PRISMA-compliant systematic review, the material selection methodology provides relevance and rigor. The referenced materials were carefully chosen with a stringent search strategy on leading academic databases and online libraries.

Databases Referenced: IEEE Xplore, ACM Digital Library, Scopus, Web of Science, PubMed, and arXiv.

Search Terms: A wide variety of keywords and their combinations were used systematically to search for relevant papers:

“fall detection” AND (“elderly” OR “older adults”)

(“wearable sensors” OR “accelerometer” OR “gyroscope” OR “inertial measurement unit” OR “IMU”) AND “fall detection”

(“deep learning” OR “convolutional neural network” OR “recurrent neural network” OR “LSTM” OR “GRU”) AND “fall detection”

“biometric data” AND “fall detection” AND (“personalization” OR “individualization”)

(“multi-modal data” OR “sensor fusion”) AND “fall detection”

“activity recognition” AND “fall classification”

Inclusion Criteria: Articles were included if they met the following criteria:

Peer-reviewed journal articles, conference publications, or quality pre-print publications.

Addressed AFDSs for the elderly.

Employed wearable sensor data (e.g., accelerometers, gyroscopes) as a major input modality.

Used machine learning or DL methods for fall classification.

Specifically examined, debated, or utilized the use of biometric information (i.e., age, gender, height, weight) in their methods or findings.

Reported empirical findings with explicit evaluation measures.

Mainly printed in the last 7 years (2018–2025), with landmark pieces prior to this timeframe included for context.

Published in English.

Exclusion Criteria: Studies were excluded if they:

Exclusively addressed vision-based FD without integration of wearable sensors.

Entirely depended on threshold-based detection methods without the use of ML/DL.

Were review articles that failed to introduce new methodologies or empirical data (although general reviews were used for background information).

Failed to have a clear explanation of their methodology, data sets, or assessment criteria.

Failed to mention FD in the elderly.

This systematic and open-ended material selection process guarantees that the cited literature creates a solid and directly applicable basis for determining the existing research landscape, identifying key knowledge gaps, and justifying our suggested multimodal DL method for improved FD.

Material and methods

We used two different models to check the correlation between biometric and sensory data.

Recurrent neural network

Recursive neural networks process sequential data and generate results based on previous calculations. The RNN input sequence as depicted in Figure 1 is translated one-to-one into the output sequence. The goal of RNN is to create a system that can generate output sequences of any length that are suitable for the current environment. RNN is used in applications such as machine translation, question answering, summarization, and dialogue modeling. RNN, like humans, can recall sequences. An RNN will predict the next scene if you ask it to. The RNN makes decisions based on both the current and prior inputs, whereas the feed-forward Neural Network simply considers the information at hand. A neural network is represented by the green

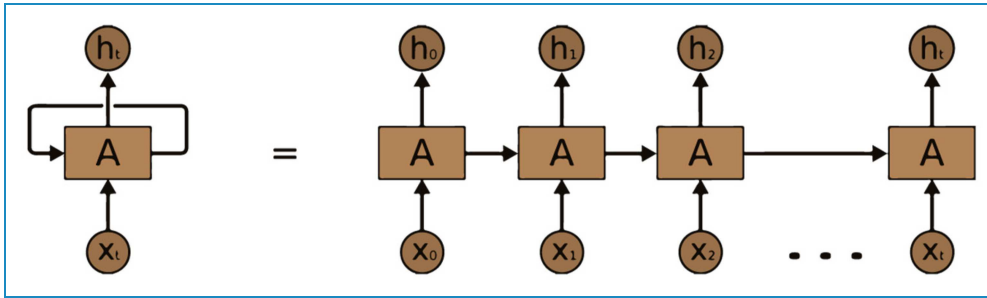


Figure 1. Recurrent neural network structure.

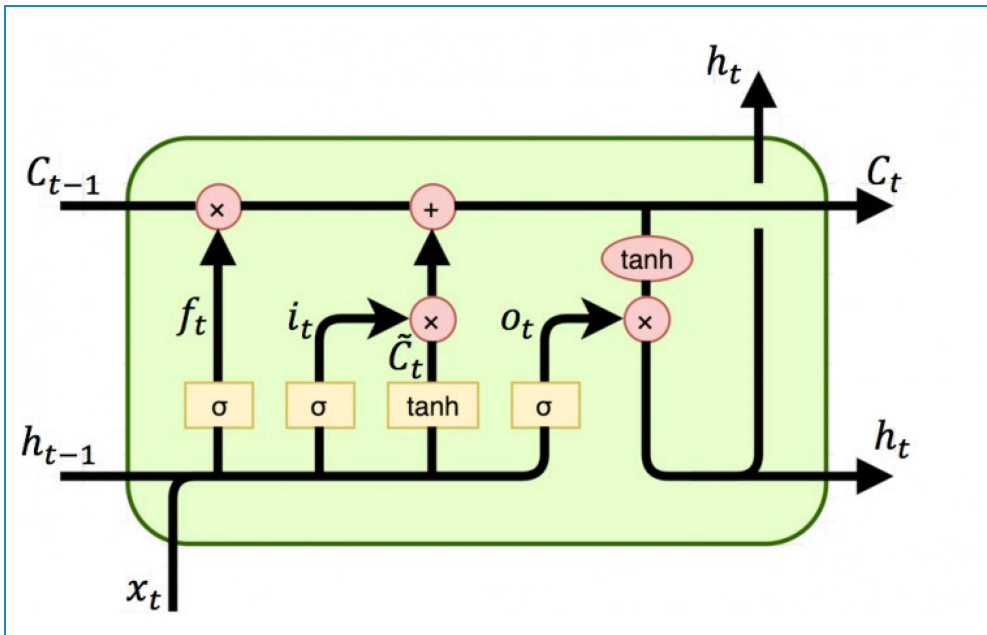


Figure 2. Long short term memory structure.

box. The arrows denote memory or merely provide feedback for the upcoming input,¹⁵ as shown in Figure 1.

LSTM network. A kind of RNN architecture titled the LSTM network depicted in Figure 2 was created specifically to solve the vanishing gradient issue that arises in conventional RNNs. A more complex structure with memory cells, input gates, forget gates, and output gates is part of its basic design. Together, these gates control how information moves throughout the network. The forget gate handles what information should be removed from the cell, the output gate chooses what information to output based on the contents of the cell, and the input gate regulates the flow of fresh information into the memory cell. LSTMs enable the network to choose store or discard information over numerous time steps, which makes them excellent at learning and maintaining long-range dependencies in sequential data. LSTMs are beneficial for a range of tasks where

retaining intricate sequential patterns is very important, including speech recognition, natural language processing, and forecasting time series. This is due to their ability to retain a long-term memory. While LSTMs may be more advanced and require greater computing resources than other versions of RNN such as GRUs, they perform wonderfully at modeling sequential data with rich temporal relationships.

The gate recurrent unit. A reduced variant of RNNs, the GRU as illustrated by Figure 3, has been developed to efficiently process sequential data. The reset gate and the update gate, which regulate information flow through the network, are the major units of its structure. The update gate decides how much of the fresh information is to be added to the current state, and the reset gate decides what past background should be forgotten. GRUs are helpful in capturing long-term dependencies in sequences since they

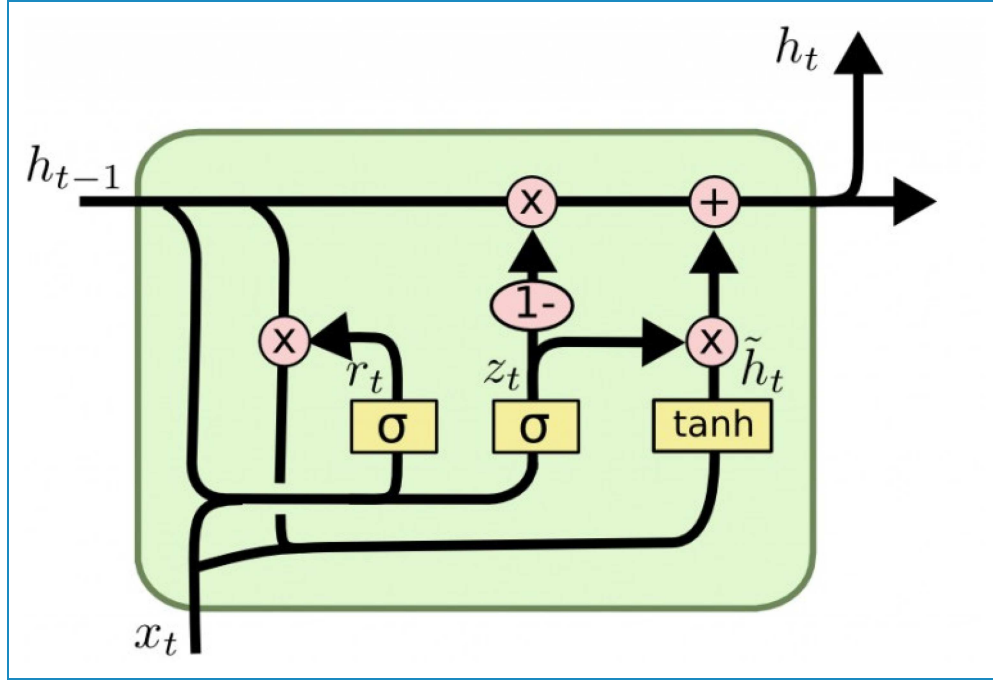


Figure 3. Gated recurrent unit structure.

can selectively forget and remember at varied time steps. Specifically, due to the fact that they have simpler structure and fewer parameters to learn, GRUs have computational benefits over LSTM networks, which are more complex. GRUs are utilized in most applications involving sequential data due to such characteristics.

Stacked LSTM model

LSTM networks, a sub-type of RNNs, are particularly good at capturing long-term dependencies in sequential data.³⁴ Geared towards addressing the vanishing gradient problem that characterizes standard RNNs, LSTM's structure involves memory cells, input, forget, and output gates to control information passage. The complex architecture allows LSTM to perform well in a range of applications requiring sophisticated sequential pattern recognition. The proposed model employs a two-layer LSTM architecture, coupled with dropout layers and a fully connected layer, to extract detailed feature representations. By mitigating the vanishing gradient issue, LSTM facilitates effective optimization and learning in deep neural networks.

- **Input Layer.** Because each activity is represented by 300 long vectors in the pre-processed dataset, the model's input dimension is set to 300.
- **LSTM Layer.** Single-layer LSTM was programmed with 512 batch sizes and a hidden number of 64 neurons. A first layer of LSTM extracts features from raw data, and a second layer classifies newly created

feature vectors. ReLU reduces over fitting by turning off a portion of the neurons' output. Because the LSTM activation function will cause instability in the model's training so Tanh activation function used instead of ReLU. Tanh is able to train the model without experiencing gradient explosion.

- **Dropout Layer.** Dropout is a regularization technique that helps neural networks generalize better by randomly deactivating neurons during training. This process reduces overfitting by preventing the network from becoming overly reliant on specific features or their interactions. Commonly used in fully connected layers with a dropout rate of 0.3, this method effectively mitigates the risk of overfitting by randomly excluding a subset of neurons in each training iteration.
- **Dense Layer.** In order to improve classification performance, the fully connected layer frequently uses the activation functions sigmoid and softmax. The sigmoid function, which is typically employed in binary classification tasks, is one of them and tends to cause the gradient disappearance problem. For this reason, in this study, we have selected the activation function Softmax.

Stacked GRU model

The GRU architecture relies on reset and update gates to regulate information flow within the network. The update gate determines the extent to which new information is

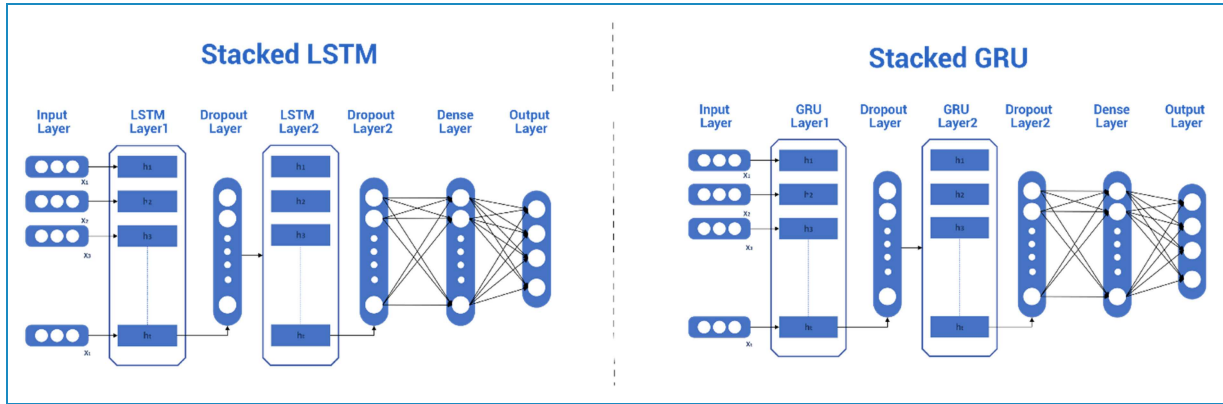


Figure 4. Model diagram for stacked GRU and stacked LSTM. GRU: gated recurrent unit; LSTM: long short-term memory.

incorporated into the hidden state, while the reset gate controls the preservation of past information. Our GRU model comprised an input layer, two GRU layers, dropout layers, a fully connected layer, and an output layer, as depicted in Figure 4. The model employed a hidden layer size of 64 and a batch size of 512, with input data dimensions matching those of the LSTM models.

Both models have been adopted from the research study. The pseudocode of the methods is detailed in Annex-A.

Dataset

This work used only the publicly accessible MobiAct dataset³⁵ for both model training and testing, formally establishing that no new participants were enrolled, thus obviating the requirement for IRB documentation. The MobiAct dataset, recorded in 2013, delivers rich information from 67 subjects (46 males, 21 females) via smartphone accelerometers, gyroscopes, and orientation sensors at 200 Hz for 12 different ADLs and simulated fall conditions. Although its public release and varied activities are major strengths, its age (2013) is still pertinent for comparison to current literature. Admitting a gender imbalance (more men) is important, as it potentially biases model performance towards male-like movements, restricting generalizability. In addition, its use of simulated falls, though controlled, necessarily does not match unpredictable real-world unplanned accidental falls, a frequent caveat in such studies. The dataset included activities mentioned in Table 1.

BSC: Falling backward when sitting on a chair, **FKL:** Falling forward when standing up, bending legs, **FOL:** Falling forward when standing up, using hands to dampen fall, **SDL:** Falling sideward when standing, first impact on knees, **CSI:** Step in a car, **CSO:** Step out of a car, **JOG:** Jogging, **JUM:** Continuous jumping, **SCH:** Sitting on a chair, **STD:** Standing with subtle, **STN:** Stairs down, **STU:** Stairs up, **WAL:** Normal walking.

Pre-processing

The DL models often require extensive data preprocessing to optimize performance on real-world datasets. Data quality directly influences learning accuracy, necessitating meticulous preparation. This process involves organizing complex data into a usable format. For this study, we employed segmentation on the original sample point sequences for each subject's acceleration, gyroscope, and biometric data. A running window of 300 sample points with a one-point overlap was applied to generate multiple data segments as proposed by Rong et al. and co-workers^{36,37} that performs its analysis exclusively on the most significant data axis, each labeled with the corresponding original sequence label.

Experiments detail

To investigate the influence of biometric data on model performance, we integrated age, gender, height, and weight into our sensory dataset. A series of experiments was conducted to evaluate the individual contributions of each biometric parameter.

A baseline model was initially trained and tested exclusively on sensory data. To assess the influence of individual biometric parameters, subsequent experiments progressively incorporated age, height, gender, and weight into the dataset. A final experiment combined all biometric data with sensory information for a comprehensive evaluation. This experimental design allowed for an examination of both individual and combined effects of biometric parameters on model performance.

Performance measures

In this study, we utilized the standard performance evaluation metrics as detailed below:

There are four types of data samples: False Positive, False Negative, True Positive, and True Negative. It is possible to compute the confusion matrix using these classification results. Sensitivity, specificity, accuracy, precision, and F1 score are the five widely used classification performance metrics in data mining, which we employ to analyze the relative performance.

3.10.1. Sensitivity (recall). Sensitivity, also known as Recall, measures the ratio of true positives that are correctly identified by the model.

$$\text{Sensitivity} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Negative}} \quad (1)$$

Specificity. It is a measure of how accurately a test identifies true negative cases. It's calculated using the following formula:

$$\text{Specificity} = \frac{\text{True Negative}}{\text{True Negative} + \text{False Positive}} \quad (2)$$

Accuracy. Accuracy is a measure of how close a measurement is to the true or accepted value. In statistical analysis, it's a measure of how often a model or test correctly predicts the outcome.

$$\text{Accuracy} = \frac{(\text{True Positive} + \text{True Negative})}{(\text{Total Number of predictions})} \quad (3)$$

Precision. In the context of statistical analysis and machine learning, it is a measure of how accurate positive predictions are. It's calculated using the following formula:

$$\text{Precision} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Positive}} \quad (4)$$

F1-score. The F1-score a harmonic mean of precision and recall, is a metric that combines precision and recall into a single score. It provides a balanced measure of a model's performance, especially when dealing with imbalanced datasets.

$$\text{F1 - Score} = 2 * \frac{(\text{Precision} * \text{Recall})}{(\text{Precision} + \text{Recall})} \quad (5)$$

Results

Falls pose a significant risk to the health and independence of older adults. While existing sensor-based FDSs are prevalent, they often neglect the impact of individual characteristics. To address this limitation, this study integrated biometric data with sensor information to enhance FD accuracy. Employing the MobiAct dataset, we trained and evaluated DL models on data collected from 57 primarily young adult participants (aged 20–47) who performed various activities and simulated falls. To assess the influence of biometric and demographic factors, we conducted six experiments progressively incorporating these features into the sensory dataset. We then applied stacked LSTM and staked GRU to detect the fall activities.

Our dataset included sensory data augmented with four biometric parameters: Age, gender, height, and weight. To assess the individual and combined impact of these parameters on model performance, we conducted a series of experiments. The baseline experiment utilized solely sensory data, while subsequent experiments progressively incorporated age, height, gender, and weight, respectively. The final experiment combined all parameters for a comprehensive evaluation.

In Table 2, Experiment 1 yielded optimal performance for both Stacked LSTM and Stacked GRU models, with the “CSO” activity achieving perfect classification accuracy (100%). The activities “FOL,” “JOG,” “JUM,” “SCH,” and “WAL” followed in descending order of performance, as detailed in Table 2. Subsequent experiments (2–6) explored the impact of different input features on model performance. While specific results are outlined in Table 2, Experiment 6, which incorporated all sensory data and additional features, demonstrated the highest overall detection accuracy. The “CSO” activity again exhibited top performance, followed by “FKL,” “JOG,” “JUM,” and other activities.

The experimental findings, compiled from both aggregate and activity-specific performance metrics in Tables 2 and 3, along with comprehensive confusion matrices, show key observations concerning how different combinations of biometrics affected the performance and misclassification rates of the Stacked LSTM and Stacked GRU models. First, the Stacked GRU model, trained only on sensory data (Experiment 1), set a strong baseline at 96% average accuracy, beating the Stacked LSTM's 93%. Both models obtained close to perfect classification for “CSO” (Class 2), reflecting its clear pattern, though early misclassifications were observed for classes such as 7 and 10, which reflects intrinsic kinematic similarities between these activities. Including age (Experiment 2) kept performance for LSTM at 93% and fell somewhat for GRU to 95%, with repeated misclassification patterns, which suggests that age introduces refined but not revolutionary contextual adaptation. In contrast, height addition (Experiment 3)

Table 1. Numbers of activities of daily living (ADL)'s and fall in the dataset.

Activity of Daily Life		Falls	
Label	Activity	Label	Activity
STD	Standing	FOL	Forward-lying
WAL	Walking	FKL	Front-knees-lying
JOG	Jogging	BSC	Back-sitting-chair
JUM	Jumping	SDL	Sideward-lying
STU	Stairs up		
STN	Stairs down		
SCH	Stand to sit(sit on chair)		
SIT	Sitting on chair		
CHU	Sit to stand(chair up)		
CSI	Car-step in		
CSO	Car-step out		
LYI	Lying		

critically impaired the Stacked LSTM performance to 58% mean accuracy with massive misclassifications of many classes (e.g., 0, 1, 5, 7, 10, 11), suggesting that concatenating height data directly injected huge noise or an uncorrelated feature space that the LSTM could not learn. The Stacked GRU was more resilient with an 88% accuracy, coping with height more elegantly. Gender inclusion (Experiment 4) sustained high performance for both models (LSTM at 92%, GRU at 95%), serving as a subtle contextual parameter that nuanced existing patterns without significantly changing kinematic signatures. Adding weight (Experiment 5) induced a modest performance reduction for both models (LSTM to 90%, GRU to 93%), indicating that weight fluctuations, when merely added, made it more difficult to distinguish subtle activity nuances. Lastly, using all biometric features together (Experiment 6) led to the largest performance degradation for the Stacked LSTM down to 84% average accuracy, showing its difficulty in working with high-dimensional, heterogeneous input and causing extensive misclassifications. The Stacked GRU, with the reduction in performance as well, was significantly more resilient at 91% average accuracy, showing its higher ability to work with multimodal data. The consistent misclassification of some activity classes (e.g., 7 and 10) in all experiments highlights that such activities present kinematic ambiguities that are beyond the capabilities of rudimentary feature extraction

or fusion methods of mere data concatenation, even for cutting-edge DL models.

The average accuracy, precision, recall, and F1-scores across experiments provide insights into the impact of biometric data on FD model performance. These metrics highlight variations in FD effectiveness among different experimental setups, emphasizing the significance of specific biometric factors.

A comparative analysis of Stacked LSTM and Stacked GRU models across various biometric and sensory data combinations is presented in the accompanying table. Each row represents a distinct experimental condition, with experiments numbered 1 to 6. The Table clearly illustrates fluctuations in F1-score, accuracy, recall, and precision across different setups. While Experiment 1, utilizing solely sensory data, achieved superior performance, Experiment 3, incorporating height with sensory data, exhibited notably lower metrics across the board. These findings underscore the importance of carefully considering biometric factors to optimize FD model performance.

Table 4 shows the p -values of a paired t -test that is aimed at differentiating the performance of six experiments using Stacked LSTM from six experiments on Stacked GRU on different classification metrics. A low p -value (generally less than 0.05) reflects that there is a statistically significant difference between the compared groups. In this table, all the p -values are significantly less than the traditional 0.05 level: Accuracy (0.0133), Precision (0.0142), Recall (0.0124), and F1-Score (0.0137). This repeated pattern of statistically significant p -values across all metrics that were tested clearly indicates that there exists a real, nonrandom difference in performance between the Stacked LSTM and Stacked GRU architectures in these experiments. Although the table strongly suggests a difference, it does not show which architecture fared better; to establish superiority, one would have to look at the actual mean performance scores for each measure for both Stacked LSTM and Stacked GRU.

These average results are also shown in the form of bar plot in Figure 2

Figure 5 reflects the results of stacked LSTM and stacked GRU which are explained in Table 3 as well. Table 3 presents a comparative analysis of Stacked LSTM and Stacked GRU models across six experiments, evaluating their performance based on average accuracy, precision, recall, and F1-score. Stacked GRU consistently outperforms Stacked LSTM across all experiments and metrics. Stacked GRU achieves an average accuracy of 93%, while Stacked LSTM's accuracy varies significantly, with a low point of 58% in Experiment 3. Both models exhibit high precision, with Stacked GRU consistently outperforming Stacked LSTM by a small margin. Similar to precision, Stacked

Table 2. Activity wise results of all experiments.

	Experiment 1						Experiment 2					
	Stacked LSTM			Stacked GRU			Stacked LSTM			Stacked GRU		
	Precision	Recall	F1-Score	Precision	Recall	F1-Score	Precision	Recall	F1-Score	Precision	Recall	F1-Score
BSC	0.89	0.93	0.91	0.93	0.97	0.95	0.89	0.95	0.92	0.91	0.96	0.94
CSI	0.90	0.94	0.92	0.97	0.95	0.96	0.94	0.92	0.93	0.94	0.94	0.94
CSO	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
FKL	0.91	0.96	0.94	0.97	0.97	0.97	0.95	0.95	0.95	0.96	0.96	0.96
FOL	0.96	0.91	0.93	0.97	0.96	0.96	0.95	0.92	0.93	0.95	0.94	0.95
JOG	0.95	0.92	0.94	0.97	0.96	0.96	0.95	0.93	0.94	0.98	0.94	0.96
JUM	0.95	0.91	0.93	0.97	0.94	0.95	0.90	0.91	0.90	0.96	0.94	0.95
SDL	0.92	0.84	0.88	0.93	0.95	0.94	0.87	0.91	0.89	0.91	0.92	0.91
SCH	0.95	0.90	0.93	0.97	0.95	0.96	0.94	0.91	0.92	0.96	0.95	0.96
STD	0.87	0.98	0.92	0.95	0.99	0.97	0.93	0.98	0.96	0.95	0.99	0.97
STN	0.91	0.90	0.90	0.94	0.96	0.95	0.94	0.92	0.93	0.94	0.95	0.94
STU	0.91	0.90	0.91	0.96	0.94	0.95	0.93	0.93	0.93	0.94	0.94	0.94
WAL	0.94	0.93	0.93	0.97	0.96	0.97	0.97	0.93	0.95	0.98	0.94	0.96
	Experiment 3						Experiment 4					
	Stacked LSTM			Stacked GRU			Stacked LSTM			Stacked GRU		
	Precision	Recall	F1- Score	Precision	Recall	F1- Score	Precision	Recall	F1- Score	Precision	Recall	F1- Score
BSC	0.41	0.66	0.50	0.78	0.90	0.48	0.90	0.93	0.91	0.90	0.96	0.93
CSI	0.93	0.39	0.48	0.91	0.77	0.83	0.96	0.92	0.94	0.97	0.95	0.96
CSO	1.00	0.99	0.99	1.00	1.00	1.00	0.99	1.00	1.00	1.00	1.00	1.00

(continued)

Table 2. Continued.

	Experiment 3						Experiment 4					
	Stacked LSTM			Stacked GRU			Stacked LSTM			Stacked GRU		
	Precision	Recall	F1- Score	Precision	Recall	F1- Score	Precision	Recall	F1- Score	Precision	Recall	F1- Score
FKL	0.73	0.80	0.76	0.89	0.94	0.91	0.98	0.92	0.95	0.97	0.97	0.97
FOL	0.62	0.77	0.69	0.90	0.88	0.89	0.93	0.94	0.93	0.96	0.96	0.96
JOG	0.71	0.47	0.57	0.91	0.89	0.90	0.97	0.91	0.94	0.98	0.94	0.96
JUM	0.53	0.69	0.60	0.86	0.90	0.88	0.94	0.91	0.93	0.94	0.94	0.94
SDL	0.38	0.30	0.34	0.90	0.85	0.87	0.85	0.91	0.88	0.96	0.92	0.94
SCH	0.42	0.36	0.39	0.89	0.79	0.84	0.87	0.87	0.87	0.94	0.95	0.94
STD	0.87	0.86	0.87	0.85	0.98	0.91	0.90	0.97	0.93	0.95	0.99	0.97
STN	0.32	0.44	0.37	0.75	0.90	0.82	0.89	0.91	0.90	0.95	0.94	0.94
STU	0.51	0.46	0.48	0.92	0.82	0.87	0.94	0.91	0.92	0.95	0.94	0.95
WAL	0.56	0.31	0.40	0.96	0.83	0.89	0.93	0.91	0.92	0.97	0.95	0.96
	Experiment 5						Experiment 6					
	Stacked LSTM			Stacked GRU			Stacked LSTM			Stacked GRU		
	Precision	Recall	F1- Score	Precision	Recall	F1- Score	Precision	Recall	F1- Score	Precision	Recall	F1- Score
BSC	0.81	0.90	0.85	0.86	0.95	0.91	0.71	0.78	0.74	0.84	0.91	0.87
CSI	0.92	0.89	0.91	0.92	0.93	0.92	0.83	0.83	0.83	0.93	0.85	0.89
CSO	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
FKL	0.94	0.90	0.92	0.96	0.94	0.95	0.93	0.84	0.88	0.97	0.85	0.90
FOL	0.94	0.89	0.91	0.94	0.94	0.94	0.80	0.91	0.85	0.81	0.94	0.87
JOG	0.93	0.90	0.92	0.96	0.91	0.94	0.91	0.81	0.86	0.95	0.91	0.93
JUM	0.88	0.89	0.88	0.92	0.93	0.92	0.89	0.79	0.84	0.94	0.86	0.90

(continued)

Table 2. Continued.

	Experiment 5						Experiment 6					
	Stacked LSTM			Stacked GRU			Stacked LSTM			Stacked GRU		
	Precision	Recall	F1 - Score	Precision	Recall	F1 - Score	Precision	Recall	F1 - Score	Precision	Recall	F1 - Score
SDL	0.85	0.85	0.85	0.90	0.91	0.90	0.73	0.82	0.77	0.86	0.94	0.89
SCH	0.85	0.89	0.87	0.92	0.89	0.90	0.83	0.77	0.80	0.87	0.90	0.89
STD	0.92	0.97	0.94	0.89	0.99	0.94	0.84	0.96	0.90	0.93	0.99	0.96
STN	0.82	0.91	0.86	0.92	0.90	0.91	0.84	0.77	0.80	0.91	0.89	0.90
STU	0.92	0.86	0.89	0.96	0.90	0.93	0.77	0.83	0.80	0.90	0.89	0.90
WAL	0.96	0.81	0.91	0.98	0.93	0.95	0.89	0.77	0.83	0.94	0.89	0.92

Experiment 1: Only sensory data, **Experiment 2:** All Sensory data + Age, **Experiment 3:** All Sensory data + Height, **Experiment 4:** All Sensory data + Gender, **Experiment 5:** All Sensory data + Weight, **Experiment 6:** All Sensory data + All Additional Features. GRU: gated recurrent unit; LSTM: long short-term memory; STD: standing; WAL: walking; JOG: jogging; JUM: jumping; STU: stairs up; STN: stairs down; SCH: stand to sit(sit on chair); SIT: Sitting on chair; CHU: sit to stand(chair up); CSI: car-step in; CSO: car-step out; LYI: lying; FOL: forward-lying; FKL: front-knees-lying; BSC: back-sitting-chair; SDI: sideward-lying.

GRU demonstrates slightly better recall across all experiments. The F1-score reinforces the overall superior performance of Stacked GRU. Experiments 1, 2, 4, and 5: Both models achieve high performance, with Stacked GRU showing a slight edge. Experiment 3: There's a significant performance drop for Stacked LSTM, indicating potential issues with the model or the data for this specific experiment. Experiment 6: While both models see a decrease in performance compared to previous experiments, Stacked GRU still maintains a higher overall score. The results strongly favor Stacked GRU over Stacked LSTM for the given task. The consistent performance of Stacked GRU across different experimental conditions highlights its robustness and effectiveness. However, the reasons for the significant performance drop in Experiment 3 for Stacked LSTM require further investigation to understand the underlying causes.

Experiment 1: Sensor data only. The first experimental phase focused exclusively on using sensory data for FD. This approach resulted in high-performance outcomes, as indicated by the key evaluation metrics.

- The **Stacked LSTM** model's accuracy was 93% which represents the percentage of correctly identified instances out of all predictions. The recall, which measures how well the model distinguishes between true and false positives, was 92%. Furthermore, a precision measure of 93% was reported, which evaluates the accuracy of positive predictions. The model achieved an F1-score of 92%, indicating a strong balance between precision and recall in identifying fall-related activities using only sensory data.

A detailed model report with recall, F1-score, and precision metrics for each activity category individually was provided in Figure 6. The model report assessed the model's overall accuracy in properly identifying instances, finding a balance between precision and recall, and recalling instances of diverse activities. Confusion matrix for this experiment is also shown in Figure 7.

- The **Stacked GRU** model demonstrated high performance in identifying fall-related activities, achieving an accuracy, recall, and precision of 96%. The F1-score of 92% further highlights the model's strong overall balance in its predictions, which were based solely on sensory values. A detailed report of the model's performance, including precision, recall, and F1-score for each activity category, is provided in Figure 8. The corresponding confusion matrix is shown in Figure 9.

Experiment 2: Sensor data + age. In the second experimentation, age is added to the sensory data as an additional

Table 3. Average results of all experiments.

	Stacked LSTM				Stacked GRU			
	Avg accuracy	Avg precision	Avg recall	Avg F1-score	Avg accuracy	Avg precision	Avg recall	Avg F1-score
Exp#1	93%	93%	92%	93%	96%	96%	96%	96%
Exp#2	93%	93%	93%	93%	95%	95%	95%	95%
Exp#3	58%	59%	58%	57%	88%	89%	88%	88%
Exp#4	92%	93%	92%	92%	95%	95%	95%	95%
Exp#5	90%	90%	90%	90%	93%	93%	93%	93%
Exp#6	84%	84%	84%	84%	91%	91%	91%	91%

GRU: gated recurrent unit; LSTM: long short-term memory

biometric variable and fed into the Stacked LSTM algorithm. The goal of this experiment is to investigate how age influences the model's functionality. By merging age information with sensory data, the goal is to see if age-related factors increase or have an impact on the model's ability to detect certain behaviors, including fall patterns. The second experimental phase resulted in a notable improvement in model performance over the initial findings.

- The accuracy of the Stacked LSTM model was 93% that translates to the proportion of instances correctly predicted out of all predictions. Recall, which quantifies how well the model separates true positives from false positives, was 93%. In addition, a precision value of 93% was recorded, which asserts the validity of positive predictions. The F1-score, harmonic mean of recall and precision that peaked at 93%, showed the balance of the model in detecting fall-related activities from only sensory values. An exhaustive model report with recall, F1-score, and precision measurements per activity category separately was given in Figure 10. The model report evaluated the model's general accuracy in correctly classifying instances, achieving a balance between precision and recall, and remembering instances of varied activities. Confusion matrix for this experiment is also displayed in Figure 11.

The **Stacked GRU** model's accuracy was 95%, recall was 95%, precision was 95%, and F1-score was 95%. A detailed report of the model's performance, including precision, recall, and F1-score for each activity category, is described in Figure 12. The corresponding confusion matrix is shown in Figure 13.

Experiment 3: Sensor data + height

- In the third experiment, the height is introduced as another biometric feature to the FD model in order to enhance the sensory data. The purpose of this experiment is to investigate the impact of an individual's height on the functionality of the model. Adding height information with sensory data is hoped to identify whether height differences influence the model's capacity for detecting various types of falls and other activities with accuracy. The accuracy of Stacked LSTM model was 58%, recall was 58%, precision was 59%, and F1-score was 57%. A model report with the recall, F1-score, and precision metrics for every activity category separately was shown in Figures 14 and 15. The model report evaluated the overall accuracy of the model in accurately detecting instances, achieving precision-recall balance, and recalling instances of varied activities. Confusion matrix for the above experiment is presented in Figure 5.
- The **Stacked GRU** model's accuracy was 88%, recall was 88%, precision was 89%, and F1-score was 88%. The per-category performance of the model is illustrated in Figure 16, which gives the recall, F1-score, and precision for every single activity. The confusion matrix for this experiment is also included in Figure 17 in order to depict the results of the classification.

Experiment 4: Sensor data + gender. Gender is added as an additional biometric feature to the FD model in the fourth experiment, enriching the sensory data. The primary goal of this experiment is to investigate any possible effects

Table 4. Significance using paired t-test to distinguish the six experiments using stacked LSTM and stacked GRU.

Metric	P-value
Accuracy	0.0133
Precision	0.0142
Re-call	0.0124
F1-score	0.0137

GRU: gated recurrent unit; LSTM: long short-term memory.

that gender may have on the model's performance. By combining gender information with sensory data, we determine whether gender-related factors affect the model's ability to correctly identify different activities, including different types of falls.

- The **Stacked LSTM** model's accuracy was 92%, recall 92%, precision was 93% and F1-score was 93%. A detailed report of the model's performance, including precision, recall, and F1-score for each activity category, is shown in Figure 18. The corresponding confusion matrix is shown in Figure 19.
- The **Stacked GRU** model's accuracy was 95%, recall 95%, precision was 95% and F1-score was 95%. Figure 20 offers a detailed breakdown of the per-category performance of the model, including the recall, F1-score, and precision measures. For additional visualization of the classification results, Figure 21 shows the confusion matrix for the current experiment.

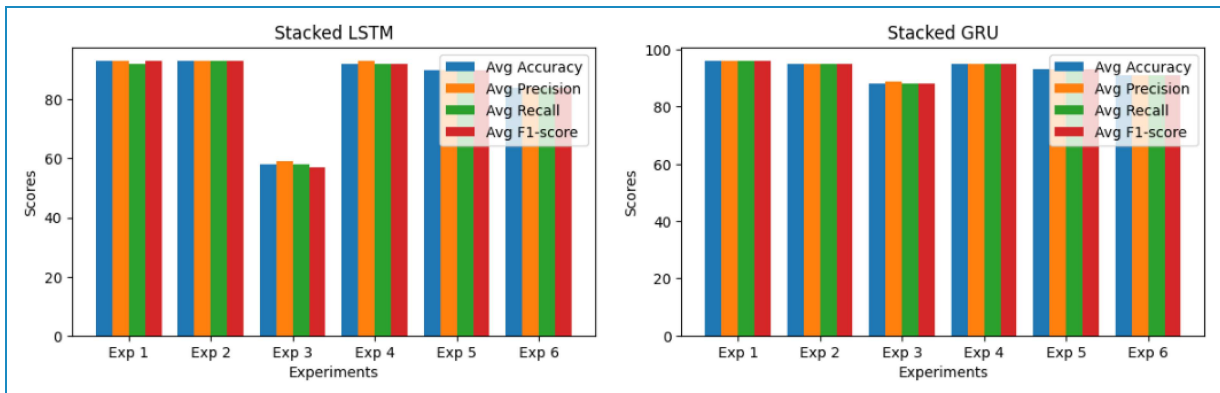
Experiment 5: Sensor data + weight. Weight is added as an additional biometric feature to the FD model in the fifth experiment, improving the sensory data. The primary goal

of this experiment is to determine whether a person's weight has an effect on the model's functionality. The goal of including weight data in addition to sensory data is to see if weight fluctuations affect the model's ability to accurately distinguish and identify different activities, such as falls.

- The **Stacked LSTM** model's accuracy was 90%, recall was 90%, precision was 90%, and F1-score was 90%. A detailed model report with recall, F1-score, and precision values for each activity category separately was depicted in Figure 22. Confusion matrix for this experiment is also depicted in Figure 23.
- The **Stacked GRU** model's accuracy was 93%, recall was 93%, precision was 93%, and F1-score was 93%. A recall-based model report along with F1-score and precision per activity class was given in Figure 24. Confusion matrix for this experiment appears in Figure 25 as well.

Experiment 6: Sensor data + biometric features. The sixth and final experiment takes a more comprehensive approach, passing this enriched dataset to the FD model after combining all available biometric features (age, height, gender, and weight) with sensory data. The primary goal of this experiment is to assess the combined effect of several biometric parameters on the model's functionality. This study evaluates the effect of biometric features (age, height, gender, and weight) on model performance. Fusing these features with sensor data, we can measure their contribution to the model's ability to correctly classify a variety of activities, such as fall events.

- The **Stacked LSTM** model's accuracy was 84%, recall was 84%, precision was 84%, and F1-score was 84%. A complete model report with recall,

**Figure 5.** Average results of both models.

	precision	recall	f1-score	support
0	0.89	0.93	0.91	113100
1	0.90	0.94	0.92	133200
2	1.00	1.00	1.00	119100
3	0.91	0.96	0.94	131100
4	0.96	0.91	0.93	118800
5	0.95	0.92	0.94	113100
6	0.95	0.91	0.93	114600
7	0.92	0.84	0.88	113700
8	0.95	0.90	0.92	122100
9	0.87	0.98	0.92	128700
10	0.91	0.90	0.90	124200
11	0.91	0.90	0.91	117900
12	0.94	0.93	0.93	99000
accuracy			0.93	1548600
macro avg	0.93	0.92	0.92	1548600
weighted avg	0.93	0.93	0.93	1548600

Figure 6. Report of sensor value only.

F1-score, and precision measurements for each activity type in isolation was presented in Figure 26. Confusion matrix for this experiment is also presented in Figure 27.

- The **Stacked GRU** model's accuracy was 91%, recall was 91%, precision was 91%, and F1-score was 91%. An in-depth model report with recall, F1-score, and precision measure for every single activity category was given in Figure 28. Confusion matrix for this experiment is also depicted in Figure 29.

The experimental findings shed light upon the performance of Stacked LSTM and Stacked GRU models under different input settings, in addition to an in-depth analysis of patterns of misclassification. In Experiment 1, using only sensory features, the Stacked LSTM model reached a general accuracy of 93%, with corresponding macro and weighted averages for precision, recall, and F1-score at 0.93 and 0.92/0.93, respectively. Its confusion matrix (Figure 7) showed high diagonal values for the majority of activities, evidencing correct classification, particularly for class 2 with precision and recall of perfect values. But misclassifications were found for class 7, frequently mixed up with classes 5, 8, 10, and 11, and for class 10, misclassified as classes 1, 5, 7, and 11. The Stacked GRU model, on the other hand, performed better in the sensory-only

setup, with an accuracy of 96% in all metrics. Its confusion matrix (Figure 9) revealed consistently high diagonal values and overall lower rates of misclassification, though residual confusions remained for class 7 into classes 5, 10, and 11.

Experiment 2 combined age as a biometric feature with sensory input. The stacked LSTM model had an accuracy of 93%, recall, precision, and F1-score all at 93%. Its confusion matrix (Figure 11) revealed equivalent patterns of misclassifications as the sensor-only LSTM for classes 7 and 10. Its Stacked GRU model, with age, achieved 95% across accuracy, recall, precision, and F1-score. The confusion matrix (Figure 13) showed good performance with slight misclassifications remaining for classes 7 and 10, a decrease from its sensor-only performance, implying that although age is applicable, its direct incorporation may introduce some slight complexities.

Experiment 3 examined the impact of height. The Stacked LSTM model had a severe performance decline, with accuracy at 58%, recall at 58%, precision at 59%, and F1-score at 57%. The confusion matrix (Figure 15) showed rampant misclassifications, especially for classes 0 and 10, to suggest that height added noise or complexity that the LSTM had difficulties learning well from. As opposed to this, the Stacked GRU model, though witnessing a dip, held up with a good performance of accuracy, recall, and F1-score of 88% and precision of 89%. Its confusion matrix (Figure 17) indicated

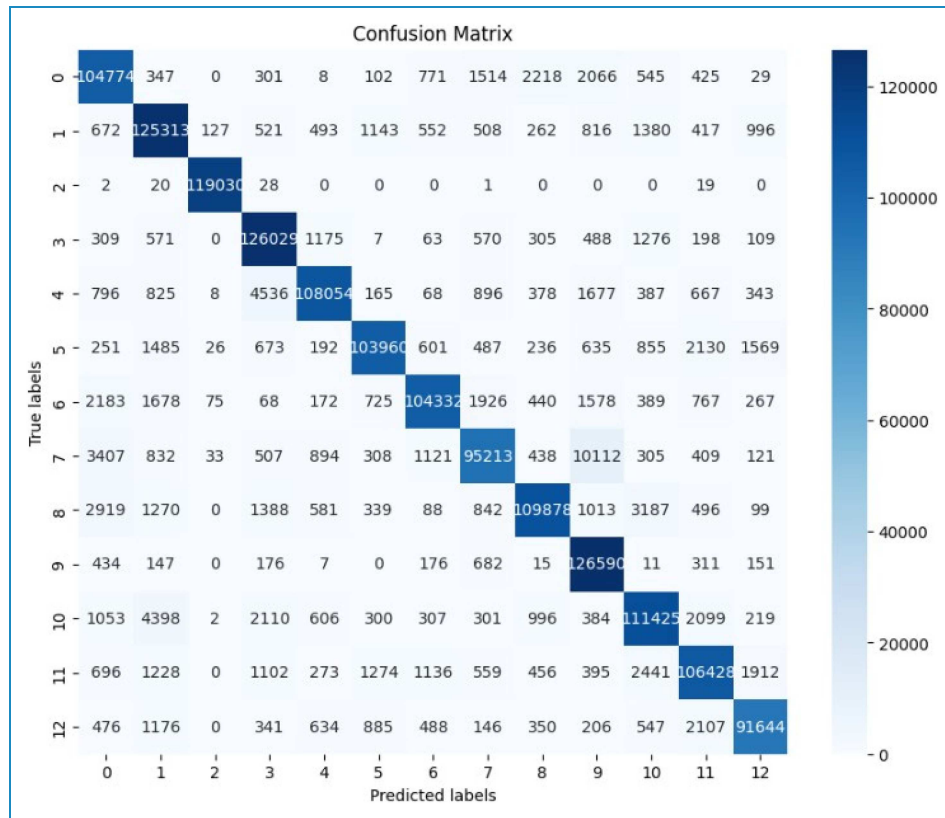


Figure 7. Confusion matrix of sensor value only.

	precision	recall	f1-score	support
0	0.93	0.97	0.95	113100
1	0.97	0.95	0.96	133200
2	1.00	1.00	1.00	119100
3	0.97	0.97	0.97	131100
4	0.97	0.96	0.96	118800
5	0.97	0.96	0.96	113100
6	0.97	0.94	0.95	114600
7	0.93	0.95	0.94	113700
8	0.97	0.95	0.96	122100
9	0.95	0.99	0.97	128700
10	0.94	0.96	0.95	124200
11	0.96	0.94	0.95	117900
12	0.97	0.96	0.97	99000
accuracy			0.96	1548600
macro avg	0.96	0.96	0.96	1548600
weighted avg	0.96	0.96	0.96	1548600

Figure 8. Report of sensor value only.

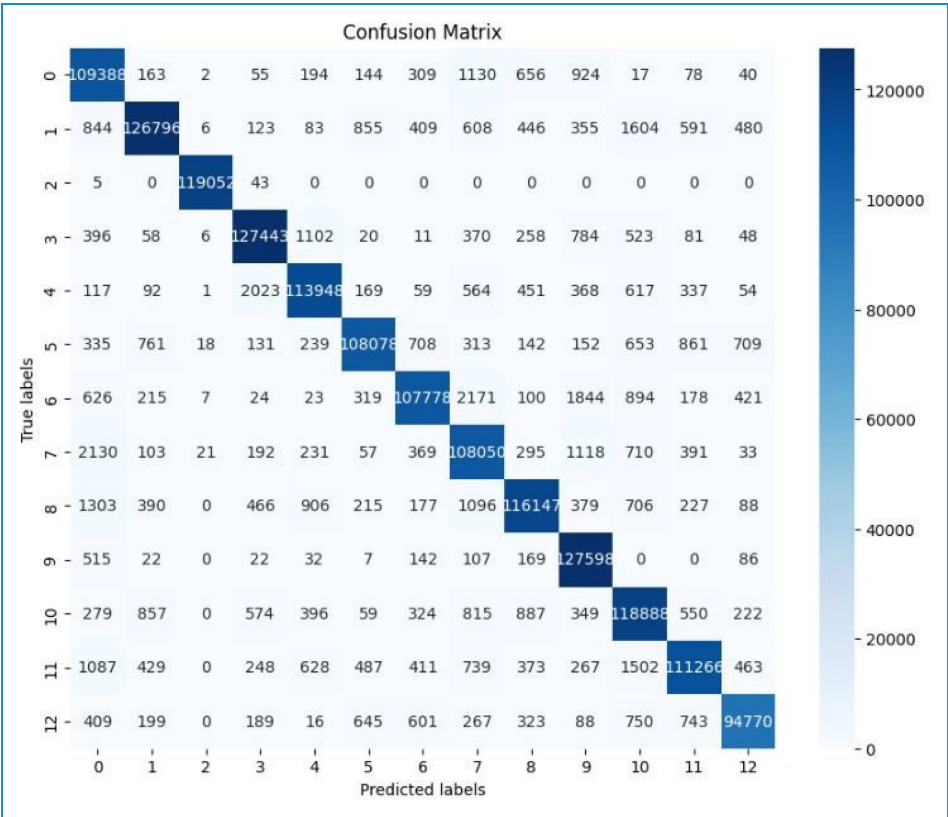


Figure 9. Confusion matrix of sensor value only.

	precision	recall	f1-score	support
0	0.89	0.95	0.92	113100
1	0.94	0.92	0.93	133200
2	1.00	1.00	1.00	119100
3	0.95	0.95	0.95	131100
4	0.95	0.92	0.93	118800
5	0.95	0.93	0.94	113100
6	0.90	0.91	0.90	114600
7	0.87	0.91	0.89	113700
8	0.94	0.91	0.92	122100
9	0.93	0.98	0.96	128700
10	0.94	0.92	0.93	124200
11	0.93	0.93	0.93	117900
12	0.97	0.93	0.95	99000
accuracy			0.93	1548600
macro avg	0.94	0.93	0.93	1548600
weighted avg	0.94	0.93	0.93	1548600

Figure 10. Report of sensor value + age.

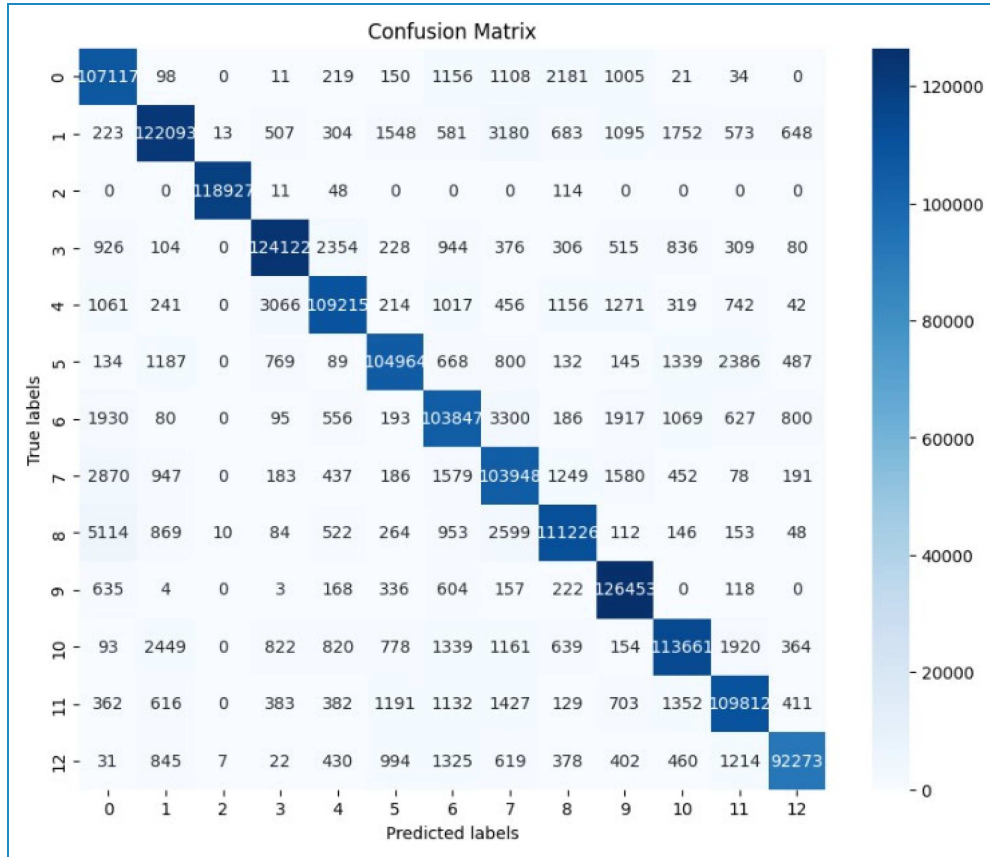


Figure 11. Confusion matrix of sensor value + age.

misclassifications, specifically for class 7 into classes 10, 11, and 4, which reflects that height may not linearly contribute to separating all activities.

Experiment 4 included gender. The Stacked LSTM model resulted in an accuracy of 92%, with recall at 92%, precision at 93%, and F1-score at 93%. Its confusion matrix (Figure 19) also reported performance similar to the sensor-only LSTM, with consistent misclassifications for class 7 and class 10. The Stacked GRU model with gender had high performance at 95% in all metrics. Its confusion matrix (Figure 21) reported good performance with minimal misclassifications for classes 7 and 10, as expected of its sensor-only and sensor + age performance.

Experiment 5 tested the effect of weight. The performance of the Stacked LSTM model dropped by a small margin, reaching 90% for accuracy, recall, precision, and F1-score. Its confusion matrix (Figure 23) had high levels of noticeable misclassifications, such as class 1 being easily mixed up with class 0, indicating difficulty in modeling weight-related variation effectively. The Stacked GRU model with weight attained 93% for all measurements. Its confusion matrix (Figure 25) indicated strong performance, although classes 7 and 10 continued to be a bit tricky.

Lastly, Experiment 6 entailed the concatenation of all biometric attributes (gender, age, height, and weight) with sensory information. The Stacked LSTM model experienced a significant degradation in performance, obtaining only 84% for all metrics. The confusion matrix (Figure 27) proved to have mass misclassifications, which hinted that raw concatenation of different features presented an inordinate amount of complexity to the LSTM. Conversely, the Stacked GRU model, after a dip from its highest, had a decent 91% in all metrics. Its confusion matrix (Figure 29) reflected better robustness than the LSTM, although class 7, 10, and 11 misclassifications remained, indicating inherent difficulties in combining such varied information.

The Stacked GRU model outperformed the Stacked LSTM in all instances, reflecting better temporal learning and data management abilities. Although single feature inclusion of age and gender typically preserved or even slightly enhanced performance, height, and most importantly the fusion of all biometrics, there were challenges. The overall systematic misclassification of activities like classes 7, 10, and 11 across models and experiments indicates that intrinsic kinematic similarities among some non-fall activities and fall events, or among various types of falls, are still a main

	precision	recall	f1-score	support
0	0.91	0.96	0.94	113100
1	0.94	0.94	0.94	133200
2	1.00	1.00	1.00	119100
3	0.96	0.96	0.96	131100
4	0.95	0.94	0.95	118800
5	0.98	0.94	0.96	113100
6	0.96	0.94	0.95	114600
7	0.91	0.92	0.91	113700
8	0.96	0.95	0.96	122100
9	0.95	0.99	0.97	128700
10	0.94	0.95	0.94	124200
11	0.94	0.94	0.94	117900
12	0.98	0.94	0.96	99000
accuracy			0.95	1548600
macro avg	0.95	0.95	0.95	1548600
weighted avg	0.95	0.95	0.95	1548600

Figure 12. Report of sensor value + age.

challenge. This is a call for more advanced feature engineering of biometric data, better fusion architectures, or attention mechanisms to be able to adequately harness multimodal inputs and minimize false alarm rates.

Discussion

The uniform dominance of the Stacked GRU model over a range of experimental settings demonstrates its robustness and flexibility with respect to contrasting data distributions. The data configuration sensitivity of the Stacked LSTM model reflects the need for meticulous feature engineering and model choice to achieve the best performance. Future research should focus on understanding the factors contributing to the performance variations observed in Experiment 3 and exploring techniques to improve the robustness of both models when dealing with complex and potentially noisy datasets. By delving deeper into these experimental outcomes, researchers can gain valuable insights into the strengths and weaknesses of different model architectures and feature engineering strategies for FD applications. The Table 5 and 6 reflects the methods based on vision based FD.

Table 4 emphasizes that vision-based FD mostly depends on the early human pose estimation and object detection methods (such as Alpha-Pose, YOLO V3, Mask

R-CNN, OpenPifPaf) to get features from video data. They are then usually inputted to DL models, specifically RNNs like LSTMs and GRUs, in order to take temporal dynamics into account, although some methods also include conventional machine learning algorithms. Although shared databases such as UR-Fall and UP-Fall are used, and assessing performance typically emphasizes accuracy, precision, recall, F1-score, and specificity, there are the usual issues of strong reliance on the accuracy of initial person detection, high computational complexity, reliance on large labeled datasets that tend to overfit, and difficulty in sustaining key-point detection post-fall.

Cai et al.³⁸ proposed a FD vision-based method based on dense blocks with a multi-channel convolution fusion strategy different from traditional Densenet. The suggested technique has the ability to eliminate data redundancy and the utilization of network parameters while also optimizing information consumption. Tested and trained it on the publicly available dataset, UR Fall, and achieved an accuracy of 95 percent. A novel human skeleton feature-based human activity recognition and FDSs has been proposed by Ramirez et al.,³⁹ in which the Alpha-Pose algorithm is used for extracting human poses in two-dimensional (2D) images from the frames extracted from video, and the fall is detected using four ML algorithms (RF, support vector machine (SVM), MLP, and KNN) by passing them these

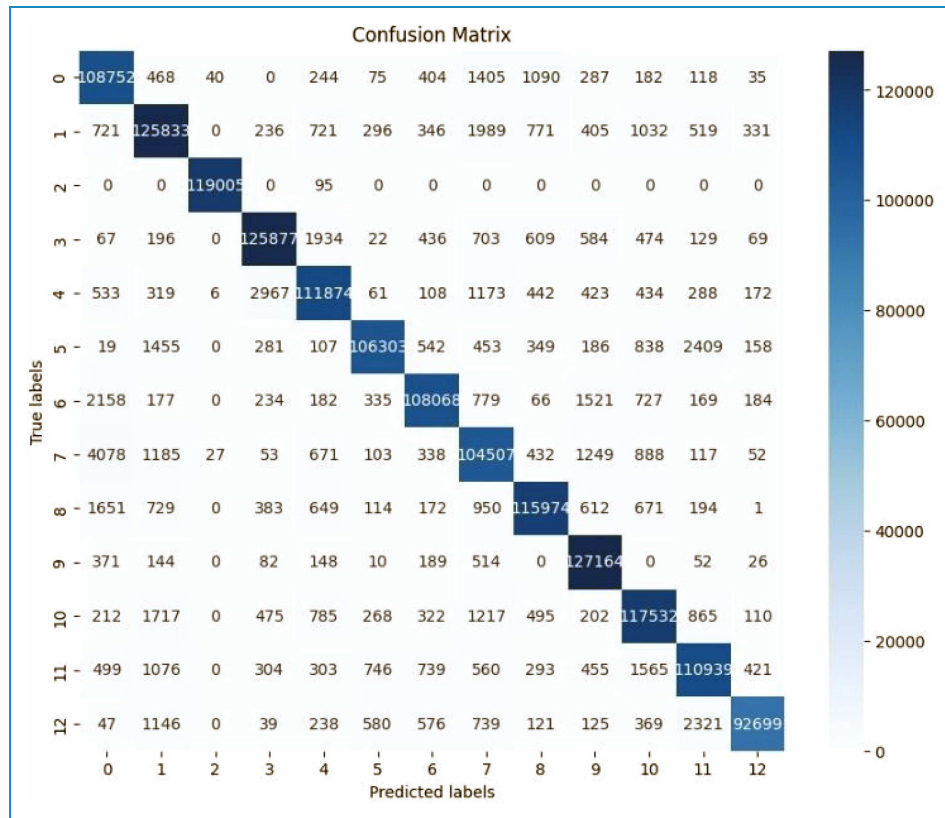


Figure 13. Confusion matrix of sensor value + age.

	precision	recall	f1-score	support
0	0.41	0.66	0.50	113100
1	0.63	0.39	0.48	133200
2	1.00	0.99	0.99	119100
3	0.73	0.80	0.76	131100
4	0.62	0.77	0.69	118800
5	0.71	0.47	0.57	113100
6	0.53	0.69	0.60	114600
7	0.38	0.30	0.34	113700
8	0.42	0.36	0.39	122100
9	0.87	0.86	0.87	128700
10	0.32	0.44	0.37	124200
11	0.51	0.46	0.48	117900
12	0.56	0.31	0.40	99000
accuracy			0.58	1548600
macro avg	0.59	0.58	0.57	1548600
weighted avg	0.60	0.58	0.58	1548600

Figure 14. Report of sensor data + height.

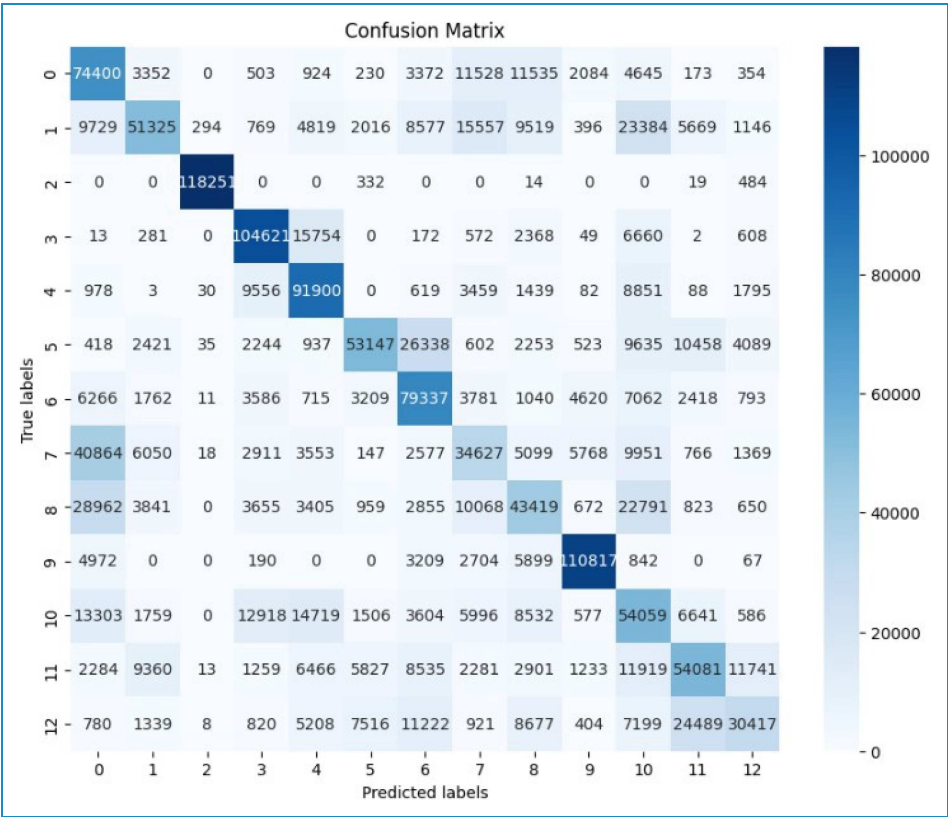


Figure 15. Confusion matrix of sensor data + height.

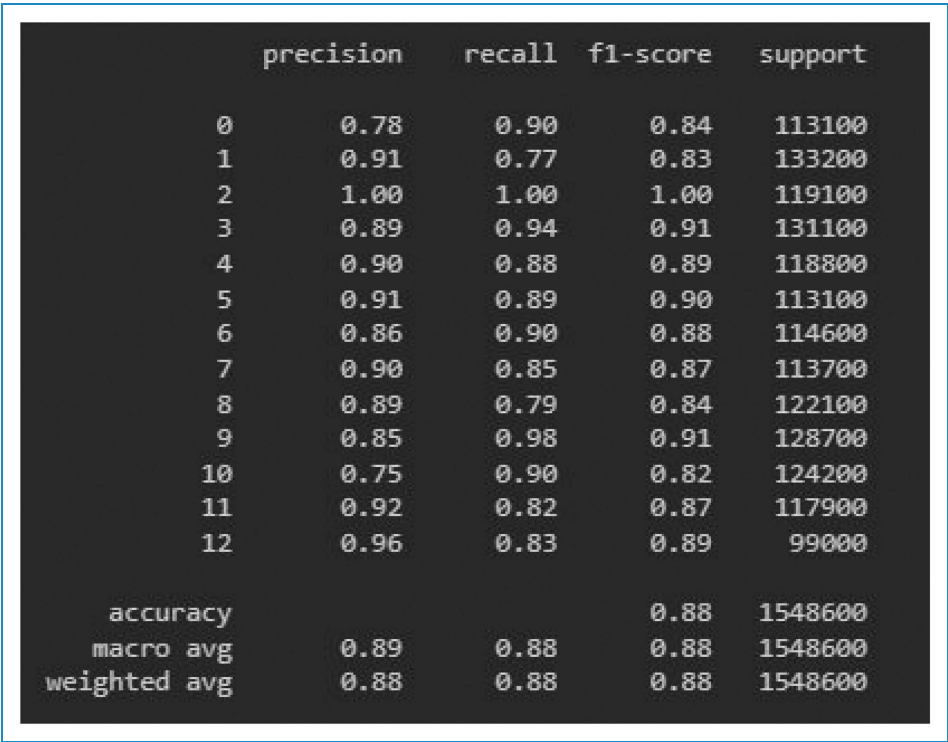


Figure 16. Report of sensor data + height.

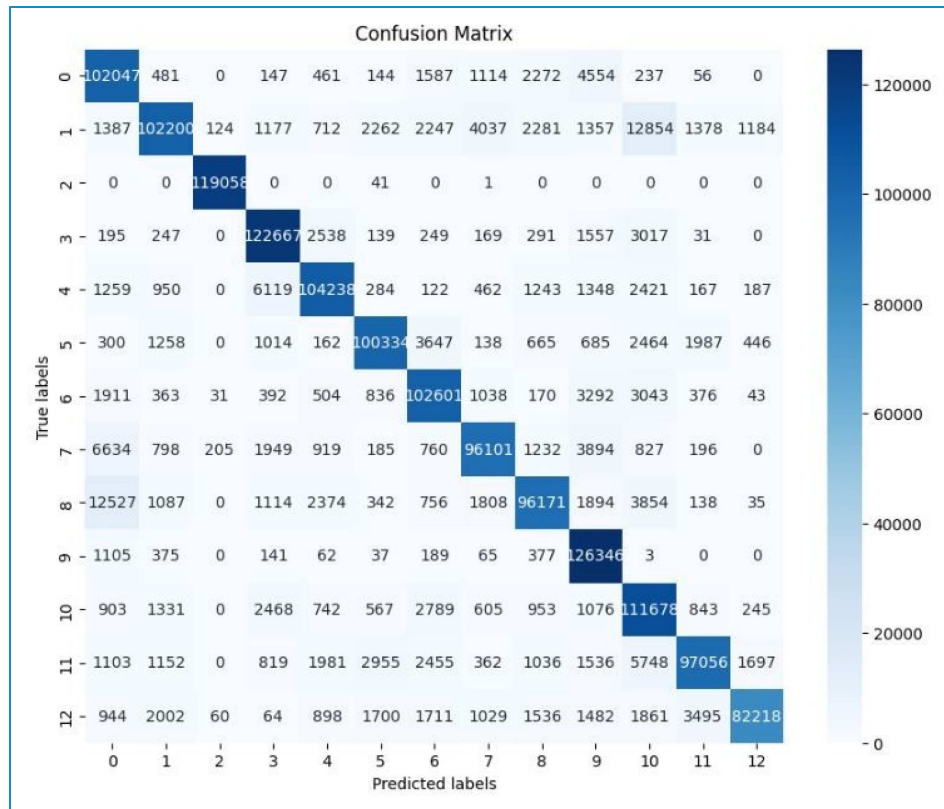


Figure 17. Confusion matrix of sensor data + height.

	precision	recall	f1-score	support
0	0.90	0.93	0.91	113100
1	0.96	0.92	0.94	133200
2	0.99	1.00	1.00	119100
3	0.98	0.92	0.95	131100
4	0.93	0.94	0.93	118800
5	0.97	0.91	0.94	113100
6	0.94	0.91	0.93	114600
7	0.85	0.91	0.88	113700
8	0.87	0.87	0.87	122100
9	0.90	0.97	0.93	128700
10	0.89	0.91	0.90	124200
11	0.94	0.91	0.92	117900
12	0.93	0.91	0.92	99000
accuracy			0.92	1548600
macro avg	0.93	0.92	0.92	1548600
weighted avg	0.93	0.92	0.92	1548600

Figure 18. Report of sensor data + gender.

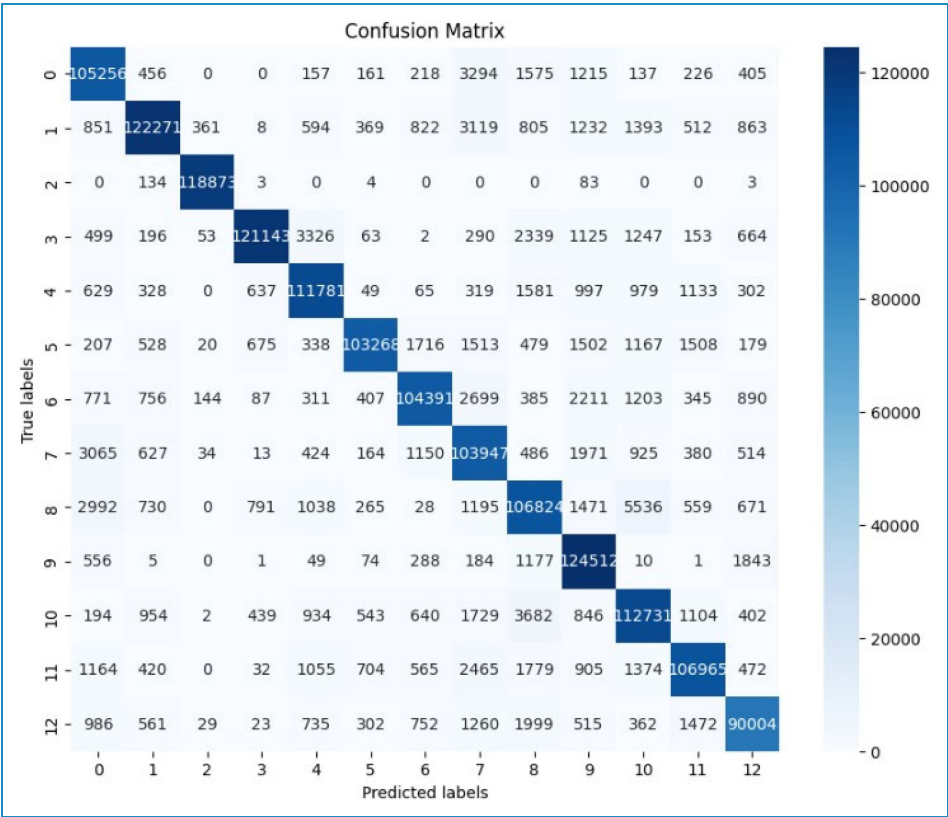


Figure 19. Confusion matrix sensor data + gender.

	precision	recall	f1-score	support
0	0.90	0.96	0.93	113100
1	0.97	0.95	0.96	133200
2	1.00	1.00	1.00	119100
3	0.97	0.97	0.97	131100
4	0.96	0.96	0.96	118800
5	0.98	0.94	0.96	113100
6	0.94	0.94	0.94	114600
7	0.96	0.92	0.94	113700
8	0.94	0.95	0.94	122100
9	0.95	0.99	0.97	128700
10	0.95	0.94	0.94	124200
11	0.95	0.94	0.95	117900
12	0.97	0.95	0.96	99000
accuracy			0.95	1548600
macro avg	0.95	0.95	0.95	1548600
weighted avg	0.95	0.95	0.95	1548600

Figure 20. Report of sensor data + gender.

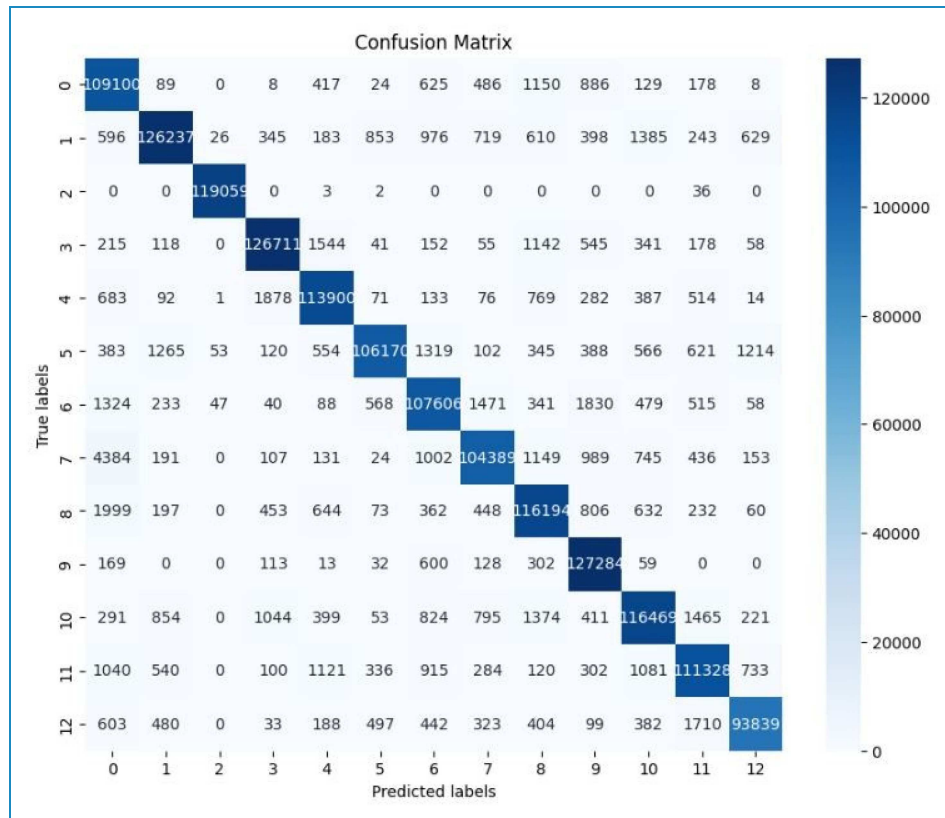


Figure 21. Confusion matrix sensor data + gender.

	precision	recall	f1-score	support
0	0.81	0.90	0.85	113100
1	0.92	0.89	0.91	133200
2	1.00	1.00	1.00	119100
3	0.94	0.90	0.92	131100
4	0.94	0.89	0.91	118800
5	0.93	0.90	0.92	113100
6	0.88	0.89	0.88	114600
7	0.85	0.85	0.85	113700
8	0.85	0.89	0.87	122100
9	0.92	0.97	0.94	128700
10	0.82	0.91	0.86	124200
11	0.92	0.86	0.89	117900
12	0.96	0.87	0.91	99000
accuracy			0.90	1548600
macro avg	0.90	0.90	0.90	1548600
weighted avg	0.90	0.90	0.90	1548600

Figure 22. Report of sensor data + weight.

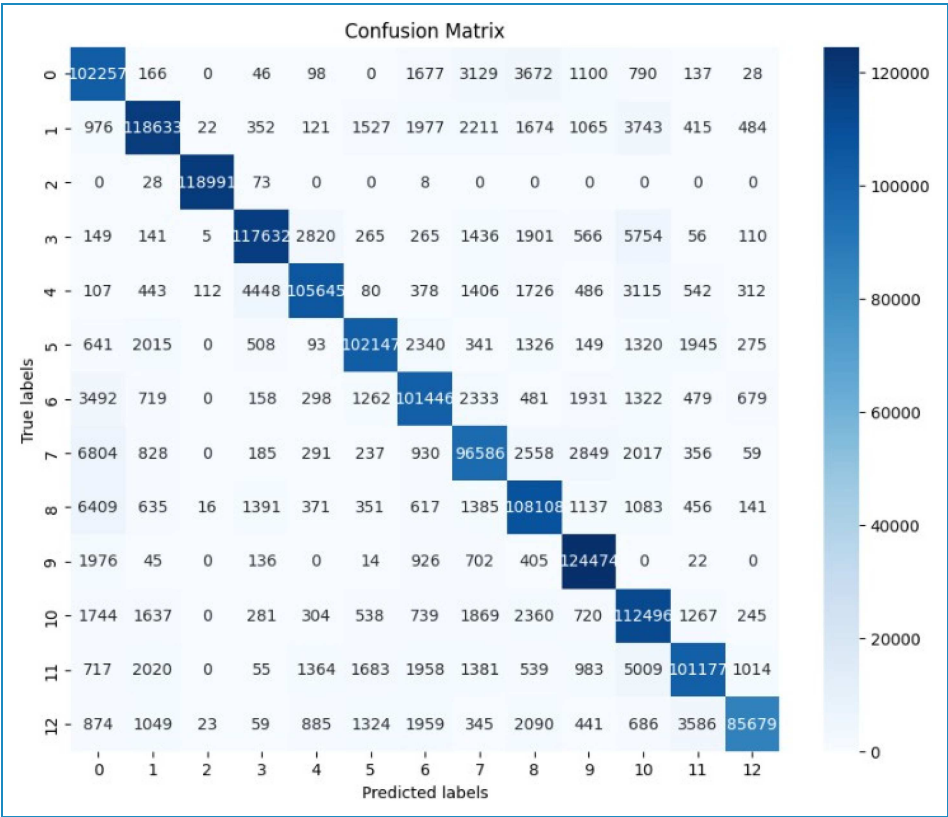


Figure 23. Confusion matrix of sensor data + weight.

	precision	recall	f1-score	support
0	0.86	0.95	0.91	113100
1	0.92	0.93	0.92	133200
2	1.00	1.00	1.00	119100
3	0.96	0.94	0.95	131100
4	0.94	0.94	0.94	118800
5	0.96	0.91	0.94	113100
6	0.92	0.93	0.92	114600
7	0.90	0.91	0.90	113700
8	0.92	0.89	0.90	122100
9	0.89	0.99	0.94	128700
10	0.92	0.90	0.91	124200
11	0.96	0.90	0.93	117900
12	0.98	0.93	0.95	99000
accuracy			0.93	1548600
macro avg	0.93	0.93	0.93	1548600
weighted avg	0.93	0.93	0.93	1548600

Figure 24. Report of sensor data + weight.

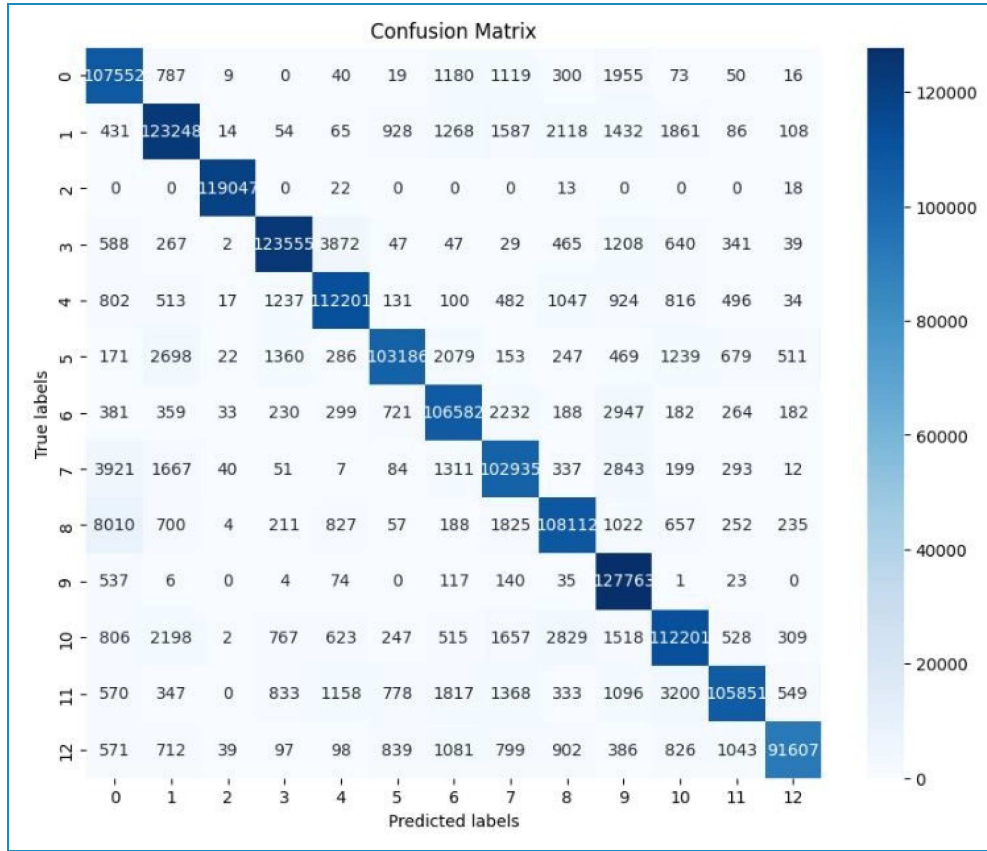


Figure 25. Confusion matrix of sensor data + weight.

2D images. For this system, use the publicly available dataset, UP-Fall. Feng et al.⁴⁰ developed an attention-guided LSTM that used spatial-temporal features for FD in a complicated environment. To begin, utilize YOLO V3 to identify pedestrians in frames, then use the Deep-Sort technique to monitor entire trajectories, extract the feature, and give it to the attention-guided LSTM. They create their own dataset for the detection approach, which consists of 220 videos (145 of which are recordings of falls). Another attention-guided bi-directional LSTM-based model is also designed for the detection of fall events in a complex environment in Chen et al.⁴¹ It consists of three parts: the mask R-CNN layer that detects people, then extracting features from that, and, at the end, passing them to the bi-direction LSTM for final prediction. They validate this model on both a publicly available dataset and their own dataset (60 fall and 180 non-fall) events. But this model depends on the accuracy of Mask R-CNN for object extraction. Salimi et al.⁴⁴ proposed a new technique based on pose estimation based on deep neural networks for human FD. In this paper, he used time distribution convolutional Long- and Short-Term Memory (TD-CNN-LSTM) and a one-dimensional CNN model to classify the data extracted from image frames as fall and not-fall. He used the UP-Fall dataset for the testing and training processes of the proposed

methodology. Another real-time FD technique based on using multiple cameras, multiple people, and LSTM was developed by Taufeeq et al.⁴² In this paper, they used the OpenPiPaf algorithm for detecting human poses from videos, then extracted five spatial and five temporal features from that data and passed it to LSTM for the final fall prediction. For validating this pipeline, they use an openly available dataset, UP-Fall. But one limitation with that pipeline is that OpenPiPaf cannot detect key points correctly once the person has fallen to the grounds. Harrou et al.⁴³ also detect human falls in a domestic setting using a vision-based approach. This approach consists of five steps: The first step is the data pre-processing step, and after that, image segmentation is performed using that data. Then features are extracted from that segmented data, and in the last fall, they were detected using a GRU chart and classified via a SVM. The datasets used for this process are the UR Fall Dataset and the FD dataset. Both are publicly available on the internet. But it takes too much time for fall prediction. Other works used vision-based data to detect potential falls using the CNN algorithm in Nunez et al. 24. According to this paper, by processing each image separately, the correlation between the frames is not considered. The optical flow algorithm addresses this issue by describing the displacement vectors between two frames. We were able to effectively represent

	precision	recall	f1-score	support
0	0.71	0.78	0.74	113100
1	0.83	0.83	0.83	133200
2	1.00	1.00	1.00	119100
3	0.93	0.84	0.88	131100
4	0.80	0.91	0.85	118800
5	0.91	0.81	0.86	113100
6	0.89	0.79	0.84	114600
7	0.73	0.82	0.77	113700
8	0.83	0.77	0.80	122100
9	0.84	0.96	0.90	128700
10	0.84	0.77	0.80	124200
11	0.77	0.83	0.80	117900
12	0.89	0.77	0.83	99000
accuracy			0.84	1548600
macro avg	0.84	0.84	0.84	1548600
weighted avg	0.84	0.84	0.84	1548600

Figure 26. Report of sensor + biometric features.

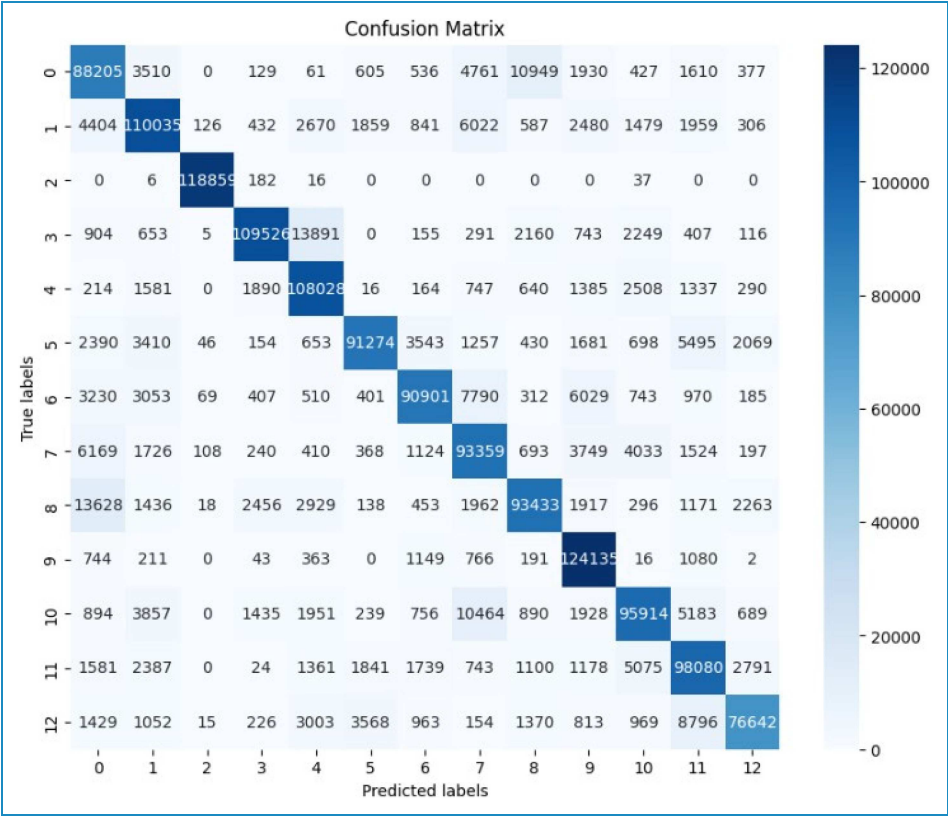


Figure 27. Confusion matrix of sensor + biometric features.

	precision	recall	f1-score	support
0	0.84	0.91	0.87	113100
1	0.93	0.85	0.89	133200
2	1.00	1.00	1.00	119100
3	0.97	0.85	0.90	131100
4	0.81	0.94	0.87	118800
5	0.95	0.91	0.93	113100
6	0.94	0.86	0.90	114600
7	0.86	0.94	0.89	113700
8	0.87	0.90	0.89	122100
9	0.93	0.99	0.96	128700
10	0.91	0.89	0.90	124200
11	0.90	0.89	0.90	117900
12	0.94	0.89	0.92	99000
accuracy			0.91	1548600
macro avg	0.91	0.91	0.91	1548600
weighted avg	0.91	0.91	0.91	1548600

Figure 28. Report of sensor + biometric features.

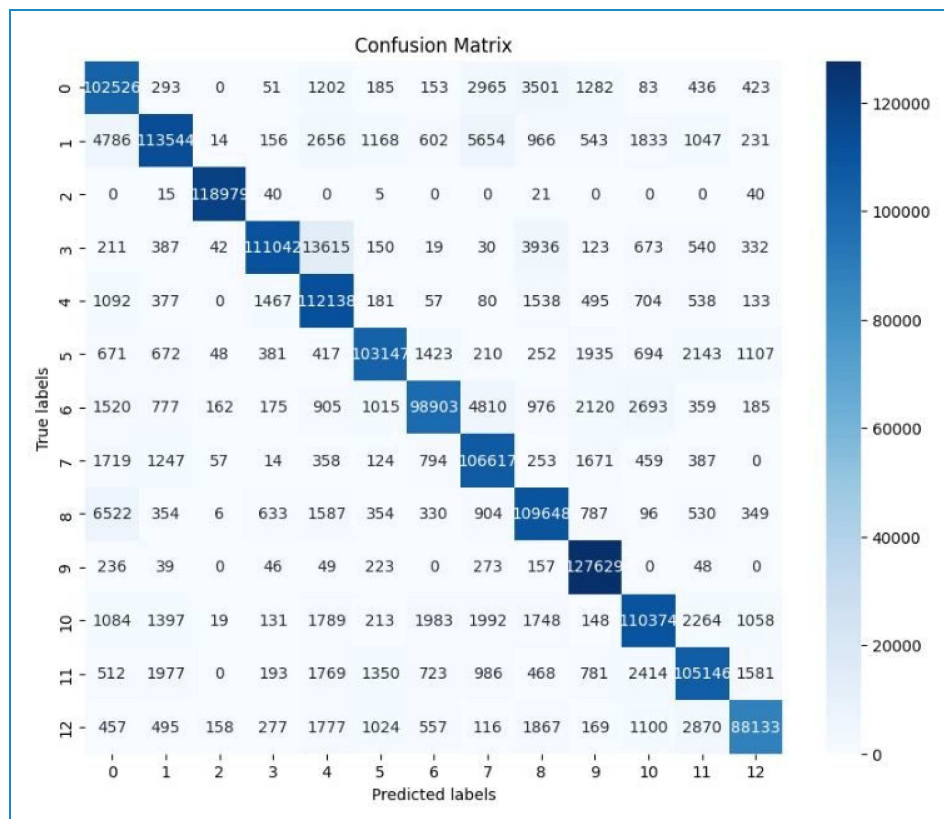


Figure 29. ConFusion matrix of sensor + biometric features.

Table 5. List of methodologies used for vision based fall detection.

Authors	Description of research	Methodology	Dataset used	Evaluation measure	Limitations
Cai et al. ³⁸	Fall detection using Multi-channel convolutional fusion strategy in dense block	The methodology of the paper involves proposing a fall detection method based on a dense block with a (MCCF) strategy.	UR-Fal	Precision, specificity, accuracy and F1-score	Need Large amount of labeled training data. Computational complexity. Over-fitting when dealing with imbalanced data.
Ramirez et al. ³⁹	Using Human Skeleton Features Fall detection and activity recognition	Firstly from videos human pose is detected using alpha-pose algorithm. Then apply 4 ML algorithm on it (RF, SVM, MLP, KNN)	UP-Fal	Accuracy, precision, specificity, F1 score.	The proposed approach does not exploit temporal consistency. Consist of 3 steps make it slower than other method.
Feng et al. ⁴⁰	Using AGLSTM, spatiotemporal fall event detection in complex settings	First use YOLO V3 to detect pedestrians then use Deep-Sort method to track complete trajectories, then extract feature and pass it to AG- LSTM.	Create own dataset of 220 videos	Precision, recall and F1-score	
Chen et al. ⁴¹	AG-Bi-LSTM-based fall event detection in complex background	It consists of three parts: Mask RCNN layer use to detect person, extract feature and then feed these feature to AG-Bi-LSTM model for fall detection.	UR fall dataset, created own dataset(60 fall +180 nonfall)	Specificity, precision, recall, F1-score, accuracy	Depend on the accuracy of Mask R-CNN method for object extraction.
Taufeeque et al. ⁴²	Real-time fall detection using multiple cameras, multiple users, and LSTM	Use OpenPiPaf algorithm for Human pose, then extract 5 temporal and spatial features. Pass these feature to LSTM for Fall detection.	UP fall dataset	Accuracy, precision, F1 score, recall	OpenPiPaf was unable to detect key points correctly once the person had fallen to the ground. small dataset
Harrou et al. ⁴³	Effective human fall detection in a home environment using a vision-based approach	Model consist of five steps: Data pre-processing, image segmentation, feature extraction, fall detection via GRU chart and classification via SVM.	UR fallFDD	Accuracy, Specificity, precision, F-measure, recall	Take too much time to predict fall event

GRU: gated recurrent unit; CNN: convolutional neural network; SVM: support vector machine; LSTM: long short-term memory; FDD: fall detection dataset.

human mobility using optical flow while avoiding the effects of static image attributes. Then, using these optical flow images, CNN was utilized to extract features, which were then passed to a fully connected neural network for the final decision of whether to fall or not.

FD technology has proven to be an important device for improving the safety and independence of older persons.

Different technologies such as wearable sensors, smart home sensors, and machine learning algorithms are used in detecting falls and sending alerts. Sensor-based solutions, which utilize wearable sensors to record body motion, are among the most common methods used. Machine learning models like LSTM, GRU, and temporal fusion transformer (TFT) are used to process time-series data from such

Table 6. List of methodologies used for wearable based fall detection.

Authors	Description of research	Methodology	Dataset used	Evaluation measure	Limitations
Kim et al. ¹⁷	Sensory data with biometric information based on multi-horizon forecasting fall detection	This paper proposed TFT-based model for fall detection using sensory information with additional biometric	DLR, Mobi-Act	F1 score, precision, recall, accuracy	Take to much time to detect Fall
Luna et al. ¹⁸	Using RNN model for fall detection from wearable sensor data.	Use single or stacked GRU for fall detection.	SiSFall	F1 score, precision, recall, accuracy	GRU is not good for long sequences
Qian et al. ¹⁹	Real-time wearable fall detection system development within the IoT context	Wearable fall detection system based algorithm, which combines MEMS with NB-IoT.	Create own dataset	Accuracy, sensitivity, specificity	MEMS sensors, when coupled with NB-IoT for constant data transmission in wearables, may drain battery life, necessitating frequent recharges
Daher et al. ²⁰	Using smart tiles, an elder tracking and fall detection system	Utilizes data force sensors with accelerometers to differentiate between falling and lying down postures	UCreating their own dataset	Accuracy, sensitivity, specificity	Too much costly.
Wu et al. ²¹	Using mobile sensors and deep learning model to detect falls	Proposes DL based model: GRUs architecture for fall detection using mobile sensors.	MobiAct, smartwatch	ASensitivity, specificity, sccuracy, precision, F1 score	GRU is not good for long sequences
Le et al. ²²	For improving the fall detection, a unique feature extraction method based on accelerometer sensor data.	Extracts 44 features based on the time domain, and frequency domain. Then tested using several ML classification: SVM, k-NN, ANN, J48, and RF.	MobiAct, UP-fall	F1-score	Take too much time to extract feature

GRU: gated recurrent unit; RNN: recurrent neural network; MEMS: micro-electromechanical system; DL: deep learning; IoT: internet of things; NB-IoT: narrow band IoT; SVM: support vector machine; TFT: temporal fusion transformer; TFT: temporal fusion transformer.

sensors. These approaches can be constrained by battery life, user wearability, and issues with data quality. Smart environment-based approaches, in contrast, use smart home appliances or dedicated sensors to sense falls. While they can provide high accuracy, these approaches can be expensive and also give rise to privacy issues.

Although the fall is known to be an abnormal activity with unique features, most of the current methods in FD are based mainly on sensor-acquired physical motion without any other information regarding the wearer of the sensor. The fast-developing area of FD does, however, highlight a move towards personalized solutions by considering wearer biometric factors like age, height, weight, and gender. The study and investigate the relationship between sensory information gathered through accelerometer and gyroscope sensors and respective biometric measures for FD. The main aim is to assess the extent to which biometric information contribute towards enhancing the accuracy and efficiency of FD model. This will lead to more personalized and effective FD methods through a deep understanding of the synergy between these heterogeneous data sources.

Table 6 provides an extensive overview of varied methodologies used in FD through wearables, from DL architectures such as TFT and GRUs to sensor-fusion methodologies utilizing MEMS and NB-IoT, and classic machine learning with thorough feature engineering. Researchers use a combination of public datasets such as MobiAct and in-house datasets, repeatedly testing performance using F1-score, Precision, Recall, and Accuracy. Critically, the “Limitations” column demonstrates typical challenges, such as inefficiency of some models with prolonged data sequences, the time-consuming character of both FD and feature extraction processes, power consumption problems in real-time IoT systems, and high cost of some specialized hardware. These discovered challenges highlight the importance of computational efficiency, real-time operation, energy minimization, and real-world affordability as research priorities for future development in wearable FD.

Table 6 provides a comparative overview of six distinct research activities in wearable-based FD, presenting a wide range of strategies from multiple authors. Methodologies vary from high-end DL architectures such as TFT and RNNs with GRU units to composite IoT systems integrating MEMS sensors with NB-IoT, conventional feature engineering in unison with machine learning classifiers (SVM, k-NN, ANN, RF), and even ambient sensing through smart tiles. Researchers utilized a combination of publicly available datasets such as MobiAct, DLR, SiSFall, and UP-Fall, and self-constructed datasets and always compared their models based on performance metrics including F1-score, Precision, Recall, Accuracy, Sensitivity, and Specificity. Nonetheless, the table also significantly points out the inbuilt limitations of each method, such as long detection latencies, difficulty in handling sequences of long data for GRUs, excessive battery usage in IoT wearables, high costs of smart tile installations for

systems, and resource-intensive feature extraction procedures. This detailed overview highlights the various avenues being pursued in FD and at the same time draws attention to ongoing difficulties in computational efficiency, power management, and real-time operation that require further research and development.

Conclusion

This work comprehensively examined the intricate interaction between single biometric features and sensory information for improved FD accuracy, using Stacked LSTM and GRU DL methods on the MobiAct data set. Our detailed analysis identified that the Stacked GRU model outperformed the Stacked LSTM in all our experiments, being more robust in processing multimodal inputs. The inclusion of single biometric attributes, like age and gender, generally preserved or slightly improved detection accuracy by offering useful contextual information. The effect of height and weight was less uniform, sometimes creating complication or even hurting performance when concatenated in isolation, especially for the Stacked LSTM.

Even using a dataset made up mainly of young adults simulating falls, our results highlight the substantial value of integrating biometric and sensor data to enhance FDSs. By doing so, it is possible to achieve a more individualized interpretation of movement patterns, which is important for segregating activities of daily living from true falls. Though the research is a preliminary step, it also underscores the paramount significance of good feature selection, sophisticated multimodal fusion methods, and well-optimized model designs in order to best utilize heterogeneous data. These findings carry important implications in designing more accurate and personalized wearable FD technology, opening the way to faster interventions in healthcare use cases. Future studies, particularly with larger, more representative datasets of older adults and actual fall environments, are critical to unlock significant discoveries and translate these into genuinely adaptive preventive interventions across the domains.

The strong performance of our Stacked GRU model, especially with the incorporation of biometric information, has important ramifications for real-world FDS development. Using individual biometric attributes such as age and gender, such systems are able to break free from a one-size-fits-all paradigm to offer personal and context-dependent monitoring. For clinical purposes, this specificity is important. As an example, a system installed within a long-term care facility may utilize a resident’s age and sex in order to more accurately distinguish between a normal movement and an actual fall, decreasing the number of false alarms that tend to be inherent within existing systems. This enhanced precision may result in more effective and trustworthy emergency response procedures, protecting healthcare professionals from being notified unnecessarily, except when an actual fall has taken place, and conserving precious time and resources. In the long run, an enhanced and customized FDS can result in quicker

interventions, mitigating the severity of fall injuries and extensively enhancing the quality of life of vulnerable patients.

Limitations and future directions

The study's focus on a predominantly young adult population limits the generalizability of findings to older adults, the primary target demographic for fall prevention systems. Additionally, the restricted ADL dataset constrains the scope of the research. While Stacked LSTM and GRU models exhibited promising results, exploring more complex architectures or ensemble methods could potentially refine model performance. Furthermore, delving deeper into feature engineering may uncover additional insights within the dataset.

To enhance the model's applicability, future research should prioritize expanding the dataset to encompass a larger, more representative sample of older adults. Incorporating a wider variety of fall types would fortify the model's robustness. Real-world testing is imperative to assess the system's practical efficacy. Investigating advanced DL techniques, such as CNNs or attention mechanisms, could potentially elevate model performance. Finally, developing a wearable device prototype would facilitate seamless integration of the FDS into users' daily lives. Moreover, we will investigate the impact of other biometric features like heart rate, blood pressure, and skin temperature. We will also combine sensor data with video or audio data to improve detection accuracy. Finally, we will utilize pre-trained models to improve performance with limited data.

The limitations of the study are that it is based on the MobiAct dataset, gathered in 2013, which would not capture contemporary movement patterns or sensor capabilities and is based on simulated falls and therefore is not generalizable to actual accidental events. The gender imbalance in the dataset and possible age biases would also distort results and influence model generalization over a wide range of populations. The straightforward concatenation approach applied for feature unification at times impaired performance, particularly for the LSTM, pointing towards an inferior fusion method. Future studies must focus on more sophisticated multimodal fusion methods (e.g., attention mechanisms) to further combine heterogeneous information, use more complete and balanced data sets that cover actual falls and more diverse physiological parameters, investigate domain adaptation to reduce the simulated-to-real gap, and improve models for resource-limited wearable device deployment for realistic, real-time implementation in elderly care.

FDSs have become an important device to promote the independence and safety of older adults. They implement different technologies, such as wearable sensors, home automation devices, and machine learning algorithms, to effectively detect falls and initiate alerts. Sensor-based approaches, which use wearable devices to record motion information, are a widely accepted method. Machine learning models,

like LSTM, GRU, and TFT, are used to classify time-series data from the sensors. Nonetheless, the approaches are restricted by battery power consumption, comfort of users, and data quality. Smart environment-based techniques, by contrast, use smart home appliances or dedicated sensors to identify falls. The approaches can provide high precision, but at a high cost and potential privacy issues.

Future studies of FD have to critically confront the weaknesses of existing research, especially those related to external validity. Based on information from young and mostly male subjects and controlled fall incidents and only from one dataset such as MobiAct, it severely limits the extension of the results to the true fall-risky elderly population and everyday life conditions. Hence, subsequent research needs to focus on enrolling heterogeneous, representative cohorts, collecting data from real fall accidents, and employing multi-dataset benchmarking to develop stronger and more universally applicable FD algorithms.

We concede that our use of statistical analysis, which only involved paired *t*-tests to distinguish between two models i.e., stacked LSTM and GRU over multiple experiments, is a limitation. Although this approach served us well for comparisons between two models in a straightforward manner, it did not enable us to get a full, simultaneous assessment of several models across multiple experimental conditions. This did not enable us to present a complete picture of the overall performance differences. For future studies, we suggest employing stronger statistical tools such as analysis of variance (ANOVA) or the Friedman test in order to compare several models at once. This will provide a stricter and more comprehensive analysis, which will make our results more valid and give a clearer picture of which models and configurations work best.

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Data availability

The MobiFall and MobiAct datasets are available upon request (please email Biomedical Informatics Laboratory at: bmi@hmu.gr) and details are publically available on <https://bmi.hmu.gr/the-mobifall-and-mobiact-datasets-2/>

Supplemental material

Supplemental material for this article is available online.

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